Cayley–Dickson Structure Census: A Computational Verification Report

From Sedenion Box-Kites to Higher-Dimensional Scaling Laws

open_gororoba Project

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Abstract

We report computational results from the `open_gororoba` research workbench concerning the algebraic structure of Cayley–Dickson algebras at successive doublings. All results are organized according to a verification ladder separating **verified algebra** (deterministic tests against known results), **statistical findings** (permutation-tested hypotheses), and **speculative models** (clearly marked). Each claim references a specific entry in the project's claims evidence matrix (C-nnn) and the corresponding Rust test function.

Contents

1 Cayley–Dickson Construction

The Cayley–Dickson doubling process Baez [2002] constructs a tower of 2^n-dimensional real algebras:

$$A_{n+1} = A_n \oplus A_n, \quad (a, b)(c, d) = (ac - \bar{d}b,\ da + b\bar{c}) \tag{1}$$

yielding the reals ($n = 0$), complex numbers ($n = 1$), quaternions ($n = 2$), octonions ($n = 3$), sedenions ($n = 4$), pathions ($n = 5$), and beyond.

At each doubling, algebraic properties degrade:

- $n = 1$: ordering lost
- $n = 2$: commutativity lost
- $n = 3$: associativity lost
- $n = 4$: alternative identity and norm-composition lost; zero divisors appear
- $n = 5$: power-associativity degrades

Our implementation uses integer-exact arithmetic for basis-element products, avoiding floating-point error entirely (claims C-001 through C-006, verified by `algebra_core::cayley_dickson` tests).

2 Zero-Divisor Census and Box-Kite Structure

At dim ≥ 16 (sedenions), zero divisors organize into combinatorial structures first described by de Marrais de Marrais [2000, 2004].

2.1 Sedenion Box-Kites

The 42 primitive cross-assessors of the sedenion algebra partition into 7 box-kites, each containing 6 assessors arranged as 3 "strut" pairs. De Marrais's published strut table matches our computational census exactly (claim C-454, test `test_strut_table_matches_de_marrais_flying_higher`).

Each box-kite's assessor interaction graph is the complete tripartite graph $K_{2,2,2}$ (octahedron). This structure is verified in `algebra_core::boxkites` (claims C-050 through C-060).

2.2 Cross-Assessor Pairs and the XOR Condition

The zero-divisor product condition for 2-blade cross-assessor pairs admits a necessary (but not sufficient) XOR filter: for a product $\alpha\beta = 0$ to hold, the basis indices must satisfy specific GF(2)-linear conditions. The XOR filter passes 315/168 pairs (ratio 0.533) with zero false negatives (claim C-445, insight I-012).

3 Motif Scaling Laws

Exact enumeration of the zero-product graph's connected components across 5 Cayley–Dickson doublings ($\dim = 16, 32, 64, 128, 256$) reveals strict scaling laws (insight I-006, claims C-126 through C-130):

Invariant	**Formula**	**Verified dims**
Components	$\dim/2 - 1$	16–256
Nodes per component	$\dim/2 - 2$	16–256
Motif classes	$\dim/16$	16–256
K_2 components	$3 + \log_2(\dim)$	16–256
K_2 part count	$\dim/4 - 1$	16–256

No octahedra ($K_{2,2,2}$) appear beyond $\dim = 16$. At $\dim = 32$, the structure completely reorganizes into 8 heptacross ($K_{2,2,2,2,2,2,2}$) and 7 mixed-degree components (insight I-006).

All motif census computations are exact (not sampled) and complete in under 2 seconds for $\dim = 256$ in release mode.

4 Lattice Codebook Filtration

Cayley–Dickson basis elements at dim $= 256, 512, 1024, 2048$ embed into an 8-dimensional integer lattice with coordinates in $\{-1, 0, 1\}$ (claims C-452, C-453, insight I-014). This embedding is dimension-independent: all four algebras use the same 8D target, suggesting the octonion sub-algebra provides the fundamental lattice structure.

Eight monograph theses (A–H) verified Reggiani [2024] (insight I-015):

Thesis A Codebook parity: coordinates sum to even, nonzero count is even, first coordinate $\neq +1$ (C-458).

Thesis B Filtration nesting: $\Lambda_{256} \subset \Lambda_{512} \subset \Lambda_{1024} \subset \Lambda_{2048}$ (C-459).

Thesis C Prefix-cut transitions: each child is a lexicographic prefix of its parent (C-460).

Thesis D Scalar shadow invariant (C-461).

Thesis E XOR partner law: partner$(i) = i \oplus (N/16)$ for $N \geq 64$ (C-462).

Thesis F Parity-clique decomposition: $K_4 \cup K_4$ at dim $= 16$, $K_8 \cup K_8$ at dim $= 32$ (C-463).

Thesis G Spectral fingerprints: $S_{\text{base}} = 2187 = 3^7$ (C-464).

Thesis H Null-model rejection (C-465).

The E8 root-system connection was tested and **refuted**: zero out of 336 pairwise lattice differences have norm-squared 2 (claim C-455, insight I-014).

5 Emanation Architecture

The de Marrais emanation construction de Marrais [2006] builds 18 layers (L1–L18) from the sedenion box-kite structure. Our implementation verifies (insight I-016, claims C-468 through C-475):

- DMZ = sign-concordance (12 edges per box-kite)
- Sail-loop = Fano incidence pattern
- Oriented Trip Sync is universal across all 7 box-kites
- 113 dedicated tests, 4400 lines of implementation

6 Ultrametric Hierarchy Fingerprint

Statistical analysis across 9 astrophysical catalogs using GPU-accelerated ultrametric fraction testing (10 million triples, 1000 permutations, RTX 4070 Ti) reveals hierarchical structure in specific physical regimes (insight I-011, I-013; claims C-436 through C-444):

Catalog	**Significant tests**	**p range**	**Verdict**
Hipparcos	48/114	< 0.05	galactic hierarchy
CHIME/FRB	8/8	< 0.01	ISM-mediated DM
ATNF Pulsars	6/8	< 0.01	ISM-mediated DM
GWOSC GW	4/66	< 0.01	mass-ratio clustering
SDSS Quasars	2/52	marginal	weak
Pantheon+ SNe	2/22	marginal	light-curve shape
McGill Magnetars	0/66	–	N too small
Fermi GBM GRBs	0/22	–	isotropic

The test discriminates physical hierarchy from noise: catalogs with known hierarchical organization (Galactic proper motions, ISM electron-density structure) show strong signal; isotropic catalogs show none (insight I-013).

7 Physical Models (Speculative Tier)

Status: speculative. The following models are exploratory and have not been validated against observational data. They are tracked in the claims evidence matrix with explicit falsification criteria.

Gravastar TOV Three-layer gravastar parameter sweep across polytropic indices (claims C-400 through C-410). Numerical solutions exist but physical relevance is untested.

Bounce cosmology Joint Pantheon+/DESI fit yields ΔBIC = +7.37 favoring ΛCDM over bounce model (claims C-200 through C-210, insight I-004). The quantum correction parameter converges to zero.

Negative-dimension eigenvalues Fractional Laplacian eigenvalue convergence under $\epsilon \to 0$ (claims C-420 through C-425).

A Claims Evidence Matrix

ID	Statement	Status	Source
C-001	Cayley-Dickson algebras become non-associative at 8D and beyond.	Verified	‘crates/algebra_core/
C-002	16D sedenions have zero divisors and lose norm composition.	Verified	‘crates/algebra_core/
C-003	”42 assessors” / ”7 box-kites” organize primitive sedenion zero divisors.	Verified	‘crates/algebra_core/
C-004	The relevant symmetry group count is ‘PSL(2,7) has order 168‘ and an explicit 168-element action is constructed which...	Verified	‘docs/SEDENION_A
C-005	”The geometry of sedenion zero divisors” (Reggiani, 2024) implies specific manifold identifications (e.g., ‘G2‘,...	Verified	‘crates/algebra_core/
C-006	GWTC-3 ”confident events” data integrated into ‘data/external/GWTC-3_confident.csv‘ and matches the GWOSC eventapi...	Verified	‘docs/archive/RESE
C-007	GWTC-3 BH mass distribution is suggestive of multimodality (Phase 2D mixture modeling); any link to \{}”negative...	Verified	‘docs/archive/RESE
C-008	Toy operator mapping: ‘alpha = -1.5‘ yields ‘abs(d_s)=2‘ under Convention B; physical interpretation remains...	Closed/Toy	‘src/scripts/analysis
C-009	Tensor-network experiment exhibits entropy scaling ‘S ˜ log(L) + Lˆ{0.5}‘.	Refuted	‘docs/archive/RESE
C-010	Hypothesis: Cayley-Dickson (sedenion) zero-divisor motifs can be mapped to physically realizable absorber networks...	Closed/Negative-Result	‘docs/MATERIALS
C-011	Hypothesis: a Cayley-Dickson (sedenion) structure could parameterize a gravastar-like effective model, but current...	Closed/Obstructed	‘docs/NAVIGATOR
C-012	”Dark Energy as Negative Dimension Diffusion” is a defensible physical interpretation (not just a metaphor).	Refuted	‘docs/NAVIGATOR
C-013	de Marrais’ GoTo ”automorphemes” (from the 7 O-trips + the ”8-ball” exclude rule) cover the 42 primitive assessors...	Verified	‘docs/DE_MARRAI
C-014	The diagonal-form family of 84 sedenion zero divisors ‘(e_low +/- e_high)‘ (from de Marrais’ 42 primitive assessors)...	Verified	‘docs/REGGIANI_R

ID	Statement	Status	Source
C-015	For each of the 84 diagonal-form zero divisors ‘u‘, there are exactly 4 other diagonal-form zero divisors ‘v‘ with...	Verified	‘docs/REGGIANI_R
C-016	The repo’s ‘m3‘ trilinear operation on distinct octonion basis triples produces exactly 42 scalar outputs and 168...	Verified	‘docs/CONVOS_CO
C-017	For diagonal 2-blades ‘(e_i +/- e_j)‘ in 16D CD, any observed zero product between two 2-blades satisfies the...	Verified	‘docs/CONVOS_CO
C-018	”Wheels” (Carlstrom) are commutative monoid-based structures ¡H,0,1,+,*,/¿ with a total reciprocal operation and...	Verified	‘docs/WHEELS_DI
C-019	Wheels (division-by-zero) provide a mathematically justified way to interpret some Cayley-Dickson zero-divisor...	Refuted	convos narrative,...
C-020	Legacy 16D ”Zero-Divisor Adjacency Matrix” represents valid algebra.	Refuted	‘data/csv/legacy/‘
C-021	1024D Basis-to-Lattice mapping is a consistent function.	Refuted	‘data/csv/legacy/‘
C-022	Toy model: map Cayley-Dickson doubling level n to surreal birthday n and test the CD property-loss milestones.	Closed/Analogy	‘docs/theory/unified
C-023	Toy model: interpret the CD associator as a discrete connection/holonomy signal over triples.	Closed/Toy	‘docs/theory/unified
C-024	C++ acceleration kernels reproduce Python CD multiplication results exactly (within float64 tolerance).	Verified	‘cpp/‘,...
C-025	GWTC-3 black hole sky positions cluster around projected sedenion zero-divisor coordinates (CMB-aligned box-kite...	Refuted	‘docs/stellar_cartogr
C-026	Speculative program: if a statistically significant ”lower mass gap” (˜2.5-5 M_sun) is established, it may correspond...	Closed/Research-Program	‘docs/stellar_cartogr
C-027	Toy model: define D_eff(rho) = 3 - k*log10(rho/rho_vac). With rho_h(M) = M/((4/3)*pi*r_s^3) at r_s = 2GM/c^2 (average...	Verified	‘docs/stellar_cartogr
C-028	Aut(S) = G2 x S3; no continuous symmetry beyond G2 emerges from sedenions. The S3 factor permutes three octonionic...	Verified	‘docs/convos/pdf_ex
C-029	Three fermion generations arise from the C tensor S decomposition into three C tensor O subalgebras (Gillard & Gresnigt...	Verified	‘docs/BIBLIOGRAI
C-030	Any sedenion-valued action/Lagrangian with products of 3+ fields must specify a bypass mechanism (explicit...	Verified	‘docs/convos/pdf_ex
C-031	By Hurwitz theorem, only dims 1/2/4/8 admit normed division algebras (R,C,H,O). In the CD tower at dim=16 (sedenions),...	Verified	‘docs/convos/pdf_ex
C-032	Tang (2025 preprint): non-associative (octonionic/sedenionic) QED uses associator norms to predict charged-lepton...	Verified	‘docs/convos/pdf_ex
C-033	Sedenion basis maps to 24 generators of SU(5) (Tang & Tang 2023).	Closed/Source-Insufficient	‘data/external/pape
C-034	Chanyal (2014): a ”sedenion unified theory of gravi-electromagnetism” expresses unified potentials/fields/currents of...	Verified	‘docs/convos/pdf_ex

ID	Statement	Status	Source
C-035	F4 Casimir ratio epsilon = C2(26)/—Delta+(F4)—= 6/24 = 1/4 exactly.	Verified	‘docs/convos/pdf_ex
C-036	Triality-governed bigraph attachment stabilizes clustering coefficient C -¿ 0.25 in thermodynamic limit.	Refuted	‘docs/convos/pdf_ex
C-037	Numerical correspondence gamma ˜ epsilon ˜ 4*lambda_GB ˜ 1/4, relating Barbero-Immirzi parameter, network clustering,...	Refuted	‘docs/C037_NUMER
C-038	Dark energy equation of state w0 = -5/6 ˜ -0.8333 emerges from twist-sector distribution in the exceptional framework.	Refuted	‘docs/convos/pdf_ex
C-039	In CDT and asymptotic safety literature, spectral dimension D_s runs from ˜4 (large scales) to ˜2 (short scales); the...	Verified	‘docs/convos/pdf_ex
C-040	Primordial tilt n_s ˜ 0.965 from fractal D_eff ˜ 2.8-3.0 at inflation.	Refuted	‘docs/convos/pdf_ex
C-041	F4 26D representation connects to bosonic string critical dimension (D=26).	Refuted	‘docs/convos/pdf_ex
C-042	Kozyrev p-adic wavelets form an explicitly computable eigenbasis for the Vladimirov operator.	Verified	‘docs/theory/PADIC
C-043	”Compact Object” populations (Pulsars, Magnetars, FRBs) can be integrated into the unified framework for...	Verified	‘docs/ROADMAP_I
C-044	Legacy 16D/32D/64D ”Zero-Divisor Adjacency Matrices” are valid basis-element maps.	Refuted	‘data/csv/legacy/‘
C-045	64-Layer Strang Splitting achieves 2nd-order convergence given finite commutator budgets.	Verified	‘docs/theory/OPER
C-046	”Fractal Doping” (sum x/nˆbeta) stabilizes zero divisors.	Refuted	‘archive/legacy_conj
C-047	E9, E10, E11 are Euclidean sphere-packing lattices.	Refuted	‘archive/legacy_conj
C-048	Analogy: ”depth stratification” is used as a terminology/structure analogy to motivic tower stratification (no physical...	Verified	‘docs/theory/OPER
C-049	Lightspace and Gravitytime are distinct geometric structures (Conformal vs Scale/Dynamics).	Established	‘docs/theory/PHAS
C-050	Toy equivalence: spaceplate delay allocation can be cast as multi-flavor flow allocation under variable renaming (LP...	Verified	‘docs/theory/WARP
C-051	A Pareto frontier exists for Spaceplates trading Compression (R) vs Bandwidth (B) vs Angle.	Verified	‘crates/optics_core/s
C-052	MERA (Multi-scale Entanglement Renormalization) circuit produces logarithmic entropy scaling S ˜ log(L).	Verified	‘src/scripts/analysis
C-053	Toy mapping: Pathion (32D) tensor diagonal -¿ dielectric stack (TMM retrieval).	Verified	‘src/scripts/analysis
C-054	Carlstrom’s Wheel Algebra formally models information loss in non-associative CD algebras.	Verified	‘crates/algebra_core/
C-055	Non-associativity is the generic (bulk) state of 16D/32D CD algebras (100% prevalence).	Verified	‘crates/algebra_core/
C-056	PDG lepton/boson masses verified against experiment (electron, muon, Z, W, Higgs).	Verified	‘src/scripts/data/fet
C-057	DESI Y1 BAO 7-bin measurements integrated into multi-probe pipeline.	Verified	‘crates/data_core/sr
C-058	Planck 2018 parameter summary + CMB spectra integrated.	Verified	‘crates/data_core/sr
C-059	NANOGrav 15yr free spectrum (log10(rho(f)) KDE) integrated.	Verified	‘crates/data_core/sr

ID	Statement	Status	Source
C-060	GWTC-3 sky localizations (64 events) integrated.	Verified	‘crates/data_core/sr
C-061	O4 GW events (10 confirmed) integrated.	Verified	‘crates/data_core/sr
C-062	CHIME/FRB 536-event catalog integrated.	Verified	‘crates/data_core/sr
C-063	ATNF 3500 pulsars + McGill 28 magnetars integrated.	Verified	‘crates/data_core/sr
C-064	Fermi GBM 3500 GRBs integrated.	Verified	‘crates/data_core/sr
C-065	CMS dimuon + diphoton spectra (J/psi, Upsilon, Z, Higgs) integrated.	Verified	‘src/scripts/data/fet
C-066	Neutrino oscillation params + KATRIN upper limit integrated.	Verified	‘src/scripts/data/fet
C-067	AFLOW 1000 + NOMAD materials + absorber experimental spectra integrated.	Verified	‘src/scripts/data/fet
C-068	Sedenion 84-ZD interaction matrix eigenvalue spectrum matches PDG particle masses.	Refuted	‘src/scripts/analysis
C-069	Three octonionic subalgebra principal angles reproduce PMNS neutrino mixing angles.	Refuted	‘src/scripts/analysis
C-070	CD associator power spectrum shape matches NANOGrav GW background.	Closed/Methodology-Insufficient	‘docs/external_sourc
C-071	FRB dispersion measures exhibit p-adic ultrametric structure.	Refuted	‘crates/stats_core/sr
C-072	CMS resonance mass ratios appear as ZD eigenvalue ratios.	Refuted	‘src/scripts/analysis
C-073	Left-multiplication operator spectrum matches PDG masses.	Refuted	‘src/scripts/analysis
C-074	**CD associator growth law: ¡—A(a,b,c)—ˆ2¿ = 2.00 * (1 - 14.6 * dˆ{-1.80}) for unit vectors.**	Verified	‘crates/stats_core/sr
C-075	**Pathion 32D ZD interaction matrix has 33 distinct eigenvalues spanning 3.3 decades.**	Verified	‘src/scripts/analysis
C-076	**Three octonionic subalgebras have exact generation symmetry (identical Casimirs, zero leakage, S3-symmetric mixing).**	Verified	‘src/scripts/analysis
C-077	Subalgebra associator mixing matrix resembles PMNS matrix.	Refuted	‘src/scripts/analysis
C-078	Higher-dim ZDs (32D/64D) improve mass spectrum coverage.	Refuted	‘src/scripts/analysis
C-079	E8 root system eigenvalues reproduce particle masses.	Refuted	‘src/scripts/analysis
C-080	FRB DM distribution has p-adic ultrametric structure.	Refuted	‘src/scripts/analysis
C-081	Multi-parameter Givens rotation finds exact PMNS angles.	Refuted	‘src/scripts/analysis
C-082	Associator saturation extends to dim 1024 with same parameters.	Verified	‘src/scripts/analysis
C-083	General-form ZDs from CSV have richer left-mult spectrum.	Refuted	‘src/scripts/analysis
C-084	Yukawa-like symmetry breaking produces PMNS-like mixing.	Refuted	‘src/scripts/analysis
C-085	CMS resonance ratios match E8+pathion combined eigenvalue ratios.	Refuted	‘src/scripts/analysis
C-086	PMNS angles from subalgebra rotation are trivially achievable.	Verified	‘src/scripts/analysis
C-087	A_inf=2 follows from statistical independence of (ab)c and a(bc).	Verified	‘src/scripts/analysis
C-088	Non-diagonal ZDs exist abundantly in 16D sedenion algebra.	Verified	‘src/scripts/analysis

ID	Statement	Status	Source
C-089	Structure constants f_{ijk} have degenerate singular values.	Verified	‘src/scripts/analysis
C-090	ZD eigenvalue spectrum is NOT invariant under SO(7) rotation.	Verified	‘src/scripts/analysis
C-091	Non-diagonal ZD spectrum matches 9/12 particle masses.	Refuted	‘src/scripts/analysis
C-092	SO(7) orbit structure of ZD spectrum is continuous, not discrete.	Refuted	‘src/scripts/analysis
C-093	Algebraic Gram matrix Tr(L_iˆT L_j) is proportional to identity.	Verified	‘src/scripts/analysis
C-094	Associator saturation fit fails for non-power-of-2 dimensions.	Verified	‘src/scripts/analysis
C-095	Associator saturation A_inf=2 is confirmed at 9 power-of-2 dims (4 through 1024) with tight error bars.	Verified	‘src/scripts/analysis
C-096	Associator tensor A(e_i,e_j,e_k) exhibits phase transitions in algebraic identities across the CD tower.	Verified	‘src/scripts/analysis
C-097	Diagonal ZD interaction graph has exactly 3 distinct edge weights {0, 1, sqrt(2)} and decomposes into 14 connected...	Verified	‘src/scripts/analysis
C-098	CD algebras lose algebraic properties at precise, dimension-specific thresholds: commutativity at dim=4, associativity...	Verified	‘src/scripts/analysis
C-099	Non-diagonal sedenion ZDs have consistent geometry: 14 nonzero components (mode), PCA dimension 13-14, kernel dimension...	Verified	‘src/scripts/analysis
C-100	Flexibility identity A(x,y,x)=0 holds exactly at all tested CD dimensions (4 through 256), while alternativity...	Verified	‘src/scripts/analysis
C-101	Flexibility identity A(x,y,x)=0 holds at ALL Cayley-Dickson dimensions through 2048.	Verified	‘src/scripts/analysis
C-102	Alternativity ratio——A(x,x,y)——ˆ2 /——A(x,y,z)——ˆ2 converges to approximately 1/2 as dim -¿ infinity.	Verified	‘src/scripts/analysis
C-103	ZD manifold topology shows sharp percolation transition at angular distance ˜1.0-1.2 radians.	Verified	‘src/scripts/analysis
C-104	Cross-term correlation decay is better modeled by inverse polynomial or log-corrected power law than pure power law.	Verified	‘src/scripts/analysis
C-105	Associator tensor SV ratio grows as dimˆ1.65, extending from dim=8 through dim=32.	Verified	‘src/scripts/analysis
C-106	Non-diagonal zero divisors exist at dim=32 with kernel dimension exactly 4 and 30 nonzero components.	Verified	‘src/scripts/analysis
C-107	Flexibility identity A(x,y,x)=0 holds to machine precision through dim=2048, with max deviations scaling as...	Verified	‘src/scripts/analysis
C-108	Alternativity ratio converges to R_inf = 0.514 +/- 0.003, consistently above 1/2 at 4.0 sigma, through dim=4096.	Verified	‘src/scripts/analysis
C-109	Random probing fails to find algebraic ZDs; diagonal ZDs lifted via CD doubling exhibit kernel doubling (kernel=8 at...	Verified	‘src/scripts/analysis
C-110	Associator tensor has multilinear rank (dim-1, dim-1, dim-1) with cubic symmetry, and CP lower bound scales as dimˆ1.10.	Verified	‘src/scripts/analysis

ID	Statement	Status	Source
C-111	Complete 13-dimension CD property table (dim=2 through 8192): flexibility and power-associativity are exactly zero at...	Verified	‘src/scripts/analysis
C-112	Alternativity ratio converges to R_inf = 0.507 +/- 0.003 with left/right symmetry, tested through dim=16384. The limit...	Verified	‘src/scripts/analysis
C-113	Moufang ratio——M(a,b,c)——ˆ2/——A(a,b,c)——ˆ2 converges to M_inf = 1.561 +/- 0.017, consistent with pi/2 (0.6 sigma),...	Verified	‘src/scripts/analysis
C-114	Full power-associativity xˆa * xˆb = xˆ{a+b} holds for all (a,b) with a+b ¡= 8 at all CD dimensions through 256, to...	Verified	‘src/scripts/analysis
C-115	The commutator [a,b] and associator A(a,b,c) are asymptotically orthogonal in high-dimensional CD algebras. The...	Verified	‘src/scripts/analysis
C-116	L-BFGS-B gradient descent finds non-diagonal ZDs at dim=16, 32, and 64, ALL with kernel dimension exactly 4. 320 ZDs...	Verified	‘src/scripts/analysis
C-117	Associator tensor spectral gap is nearly constant (˜3.0) while effective rank grows as dimˆ0.80. Entropy grows...	Verified	‘src/scripts/analysis
C-118	The Jordan identity J(xˆ2, y, x) = (xˆ2*y)*x - xˆ2*(y*x) = 0 holds at ALL Cayley-Dickson dimensions from 2 through...	Verified	‘src/scripts/analysis
C-119	The left and right Bol identities hold exactly through dim=8 (octonions) and are lost at dim=16 (sedenions), with L/R...	Verified	‘src/scripts/analysis
C-120	Diagonal ZD kernels scale as dim/4 (not universally 4). Lifted 16D ZDs also have kernel = dim/4. Kernel universality...	Verified	‘src/scripts/analysis
C-121	Multi-seed bootstrap: Moufang ratio converges to M_inf = 1.519 +/- 0.013, consistent with 3/2 (1.5 sigma), INCONSISTENT...	Verified	‘src/scripts/analysis
C-122	Multi-seed bootstrap: Alternativity ratio converges to R_inf = 0.504 +/- 0.002, consistent with 1/2 (1.9 sigma).	Verified	‘src/scripts/analysis
C-123	The associator Lie bracket [A(a,b,c), A(d,e,f)] = A1*A2 - A2*A1 has relative norm stabilizing at ˜1.97 and is...	Verified	‘src/scripts/analysis
C-124	The flexibility identity (ab)(ca) = a((bc)a) holds exactly through dim=8 (octonions) and is lost at dim=16 (sedenions),...	Verified	‘src/scripts/analysis
C-125	Artin’s theorem (2-generated subalgebras are associative) holds through dim=8 and fails at dim=16. However, flexibility...	Verified	‘src/scripts/analysis
C-126	The nested Lie bracket [[A1,A2],A3] has relative norm stabilizing at ˜1.95 (range 0.054 at dim ¿= 32). The Jacobi...	Verified	‘src/scripts/analysis
C-127	The full Jordan product x o y = (xy+yx)/2 satisfies the Jordan identity (x o y) o (x o x) = x o (y o (x o x)) at ALL CD...	Verified	‘src/scripts/analysis
C-128	The conjugate inverse xˆ{-1} = conj(x)/——x——ˆ2 gives x * xˆ{-1} = xˆ{-1} * x = 1 at ALL CD dimensions through 256, to...	Verified	‘src/scripts/analysis

ID	Statement	Status	Source
C-129	The associator norm distribution concentrates as dim grows (CV -¿ 0.01). The distribution has slight positive skew...	Verified	‘src/scripts/analysis
C-130	The associator norm——A(a,b,c)——-¿ sqrt(2) because (ab)c and a(bc) become uncorrelated (cos -¿ 0) while both preserve...	Verified	‘src/scripts/analysis
C-131	Identity violation ratios are universal constants: alt/assoc -¿ 1/2, mouf/assoc -¿ 3/2, flex/assoc -¿ 3/2. Moufang and...	Verified	‘src/scripts/analysis
C-132	The commutator norm——[a,b]——ˆ2 -¿ 4.01 (——[a,b]——-¿ 2.0). Commutator is zero at dim=2 (commutative) and nonzero...	Verified	‘src/scripts/analysis
C-133	The Moufang defect and associator are asymptotically orthogonal (cos -¿ 0, perp_frac -¿ 1.0). This parallels the...	Verified	‘src/scripts/analysis
C-134	ZD pair products at dim=16 show rich combinatorial structure: of 18 tested pairs, 4 give u*v=0 (mutual annihilation), 7...	Verified	‘src/scripts/analysis
C-135	Power norms——xˆn——= 1 EXACTLY (to machine precision) at ALL CD dimensions through 256 for ALL powers through xˆ16....	Verified	‘src/scripts/analysis
C-136	Norm multiplicativity——xy——=——x——*——y——holds exactly through dim=8 (composition algebras) and is lost at dim=16; mean...	Verified	‘src/scripts/analysis
C-137	ZD products at dim=32 preserve the norm trichotomy {0, 1, sqrt(2)} from dim=16, but kernel diversity increases: kernels...	Verified	‘src/scripts/analysis
C-138	3-generated subalgebras become non-associative at dim=8 (octonions), while 2-generated subalgebras (Artin’s theorem)...	Verified	‘src/scripts/analysis
C-139	Violation ratios at dim=8192 with 5-seed bootstrap: alt/assoc = 0.497 +/- 0.002 (1.7 sigma from 1/2), mouf/assoc =...	Verified	‘src/scripts/analysis
C-140	Associator component entropy: relative entropy ˜ 0.84 (not uniform), effective dimension ˜ 0.48 * dim (about half of...	Verified	‘src/scripts/analysis
C-141	Mixed product norms:——(ab)c——and——a(bc)——both concentrate near 1.0 for unit inputs. The ratio——(ab)c——/——a(bc)——-¿ 1 as...	Verified	‘src/scripts/analysis
C-142	Power-associativity xˆm * xˆn = xˆ(m+n) holds exactly (machine epsilon) at ALL CD dimensions through 512, for all...	Verified	‘src/scripts/analysis
C-143	Left and right multiplication operators L_a, R_a have identical singular value spectra (SV overlap = 1.0) and are...	Verified	‘src/scripts/analysis
C-144	ZD kernel spectrum at dim=64 has 9 distinct values {4, 12, 16, 20, 24, 28, 32, 36, 40}, all multiples of 4. This is...	Verified	‘src/scripts/analysis
C-145	Four-element products have exactly 5 distinct bracketings. At dim=4 (associative), all are identical (0 distinct...	Verified	‘src/scripts/analysis
C-146	Inner derivation D(a,b)(x) = [[a,b],x] + [[a,x],b] + [[x,b],a] satisfies the Leibniz rule D(xy) = D(x)y + xD(y) exactly...	Verified	‘src/scripts/analysis

ID	Statement	Status	Source
C-147	Alternator-associator decomposition: A(a,b,c) splits into alternating part Alt = A(a,b,c) - A(b,a,c) and symmetric part...	Verified	‘src/scripts/analysis
C-148	The nucleus N(A) of a CD algebra equals the full algebra at dim=4 (quaternions are associative) and equals only the...	Verified	‘src/scripts/analysis
C-149	Composition defect delta =——xy——ˆ2 - ——x——ˆ2*——y——ˆ2 is identically zero at dim ¡= 8 (Hurwitz theorem). At dim ¿= 16,...	Verified	‘src/scripts/analysis
C-150	Quadruple associator A(a,b,c,d) has——A(a,b,cd)——˜——A(a,b,c)——(ratio 0.8-1.2) at all dims. The derivation-like fraction...	Verified	‘src/scripts/analysis
C-151	At dim=64, ALL 780 pairwise products of 40 diagonal-form ZDs produce non-ZD elements with norm exactly 1.0. No zero...	Verified	‘src/scripts/analysis
C-152	Bracket polynomial: at dim=4 (associative), all bracketings of n-ary products are identical (1 distinct). At dim ¿= 8,...	Verified	‘src/scripts/analysis
C-153	Conjugate associator identity: A(conj(a),conj(b),conj(c)) = conj(A(a,b,c)) = -conj(A(c,b,a)) holds EXACTLY (machine...	Verified	‘src/scripts/analysis
C-154	Artin’s theorem: any 2-generated subalgebra is associative at dim=4 and dim=8 (alternative algebras). Fails at dim ¿=...	Verified	‘src/scripts/analysis
C-155	Jordan identity (xy)xˆ2 = x(yxˆ2) holds EXACTLY (machine epsilon) at ALL CD dimensions through 256. This is a universal...	Verified	‘src/scripts/analysis
C-156	ZD interaction graph (edge iff product = 0): dim=16 has 84 ZDs, 168 edges, 7 components, 4-regular (uniform degree 4)....	Verified	‘src/scripts/analysis
C-157	Jacobi identity [[a,b],c] + [[b,c],a] + [[c,a],b] = 0 holds exactly at dim=2 (commutative, all commutators zero) and...	Verified	‘src/scripts/analysis
C-158	Idempotent structure: the only idempotent in CD algebras is e_0 (the identity element). No nontrivial idempotents (eˆ2...	Verified	‘src/scripts/analysis
C-159	Trace form Tr(x) = 2*Re(x) is bilinear (additive + homogeneous), and the inner product ¡x,y¿ = Re(conj(x)*y) is...	Verified	‘src/scripts/analysis
C-160	All three Moufang identities (left, right, middle) hold exactly at dim ¡= 8 and fail together at dim ¿= 16. The defect...	Verified	‘src/scripts/analysis
C-161	The associator is multilinear (trilinear): A(alpha*a,b,c) = alpha*A(a,b,c) and A(a+a’,b,c) = A(a,b,c) + A(a’,b,c), in...	Verified	‘src/scripts/analysis
C-162	ZD annihilator dimensions: at dim=16, diagonal-form ZDs have left/right annihilator dim in {0, 4}. At dim=32, values in...	Verified	‘src/scripts/analysis
C-163	Commutator-associator cross-structure:——[A(a,b,c), d]——/——A(a,b,c)——converges to ˜2.0 at high dim....	Verified	‘src/scripts/analysis
C-164	n-fold product norms: at composition dimensions (dim ¡= 8),——a1*a2*...*an——= 1.0 exactly for ALL n up to 20. At dim ¿=...	Verified	‘src/scripts/analysis

ID	Statement	Status	Source
C-165	Generic SO(dim) rotations do NOT preserve associator norms. Relative difference——A_orig—	Verified	‘src/scripts/analysis
C-166	The flexible identity (ab)a = a(ba) holds EXACTLY (machine epsilon) at ALL CD dimensions through 512. This is...	Verified	‘src/scripts/analysis
C-167	Left alternative A(a,a,b) = 0 and right alternative A(a,b,b) = 0 hold exactly at dim ¡= 8 and fail together at dim ¿=...	Verified	‘src/scripts/analysis
C-168	ZD product spectrum at dim=128: among the first 100 diagonal-form (e_i+e_j)/sqrt(2) candidates, ZERO are actual zero...	Verified	‘src/scripts/analysis
C-169	Associator trilinearity via commutator: A(a,b,[c,d]) = A(a,b,cd) - A(a,b,dc) holds EXACTLY at ALL dims. This is a...	Verified	‘src/scripts/analysis
C-170	Associator norm——A(a,b,c)——for random unit vectors converges to sqrt(2) from below as dim increases. Deviation from...	Verified	‘src/scripts/analysis
C-171	Left multiplication operator determinant:—det(L_a)—= 1 exactly for unit a at composition dimensions (dim ¡= 8). At...	Verified	‘src/scripts/analysis
C-172	Center Z(A) = full algebra at dim=1,2 (commutative); Z(A) = R*e_0 (dim=1) at dim ¿= 4. The commutator map z -¿ [z, e_i]...	Verified	‘src/scripts/analysis
C-173	Power norm ratio——x^n——/——x——^n = 1.0 EXACTLY at ALL CD dimensions through 128, for all n from 2 to 10. This is a...	Verified	‘src/scripts/analysis
C-174	Left-right operator commutant [L_a, R_b] = 0 exactly at associative dimensions (dim ¡= 4). At non-associative dims (¿=...	Verified	‘src/scripts/analysis
C-175	Associator subspace Assoc(A) = span{A(a,b,c)} has rank = dim - 1 (= pure imaginary part) at all tested dims (8-64). It...	Verified	‘src/scripts/analysis
C-176	Commutator/product norm ratio——[a,b]——/——ab——converges to 2.0 from below as dim increases: 1.14 (dim=4), 1.60 (dim=8),...	Verified	‘src/scripts/analysis
C-177	Two-generated subalgebra dimension: at dim=4 (quaternions), ¡a,b¿ = full algebra (dim=4). At dim=8, median sub_dim = 4...	Verified	‘src/scripts/analysis
C-178	Inner derivation space dimension (via D(a,b)(c) = A(a,b,c) - A(b,a,c) + A(c,a,b)): dim=0 at dim=2,4 (associative -¿...	Verified	‘src/scripts/analysis
C-179	No nonzero nilpotent elements exist in any CD algebra. For unit vectors,——x^n——= 1.0 exactly for all n tested (2...	Verified	‘src/scripts/analysis
C-180	Inner product preservation ¡ax,ay¿ =——a——^2 ¡x,y¿ holds EXACTLY at composition dimensions (dim ¡= 8) and FAILS at dim...	Verified	‘src/scripts/analysis
C-181	Quadratic representation U_a(x) = 2a(ax) - (a^2)x is NOT an algebra endomorphism at any CD dimension. Relative...	Verified	‘src/scripts/analysis
C-182	Associator alternation: A(a,b,c) + A(b,a,c) = 0 (skew-symmetry in first two slots) holds EXACTLY at dim ¡= 8...	Verified	‘src/scripts/analysis

ID	Statement	Status	Source
C-183	Iterated commutator:——[[a,b],c]——/——[a,b]——- ¿ 2.0 as dim increases (1.34 at dim=4, 1.98 at dim=256). Jacobi identity...	Verified	‘src/scripts/analysis
C-184	Malcev identity [J(a,b,c), a] + J(a, b, [c,a]) = 0 holds EXACTLY at dim=8 (with opposite sign convention: LHS = -RHS)....	Verified	‘src/scripts/analysis
C-185	Product norm distribution:——ab——= 1.0 exactly at composition dims (¡=8). Beyond,——ab——˜ 1.0 +/- sigma where sigma...	Verified	‘src/scripts/analysis
C-186	Nested associator (triassociator):——A(A(a,b,c),d,e)——/——A(a,b,c)——˜ 1.2-1.5 at all tested dims (8-256). The nesting...	Verified	‘src/scripts/analysis
C-187	Every nonzero CD element has a two-sided inverse: a * conj(a)/——a——ˆ2 = conj(a)/——a——ˆ2 * a = e_0 EXACTLY at ALL...	Verified	‘src/scripts/analysis
C-188	Quaternionic subalgebra {e_0,e_1,e_2,e_3} is ASSOCIATIVE inside all higher-dimensional CD algebras (dim=8,16,32,64)....	Verified	‘src/scripts/analysis
C-189	Associator map T_{a,b}: c -¿ A(a,b,c) has purely imaginary eigenvalues (skew-symmetric) at dim ¡= 8 (alternative...	Verified	‘src/scripts/analysis
C-190	Bol identity a(b(ac)) = (a(ba))c holds EXACTLY at dim ¡= 8 (alternative algebras). Fails at dim ¿= 16 with increasing...	Verified	‘src/scripts/analysis
C-191	Double commutator norm ratio——[a,[b,c]]——/——[b,c]——approaches 2.0 monotonically from 1.35 (dim=4) through 1.99...	Verified	‘src/scripts/analysis
C-192	CD doubling formula: our cd_multiply_batch does NOT use the standard Cayley-Dickson doubling formula (a,b)(c,d) = (ac -...	Verified	‘src/scripts/analysis
C-193	Conjugate reversal conj(ab) = conj(b)*conj(a) holds EXACTLY (max_diff = 0.0) at ALL CD dimensions from 2 through 512....	Verified	‘src/scripts/analysis
C-194	Four-element associator: five parenthesizations of abcd all produce equal norms at dim ¡= 4 (associative). At dim ¿= 8,...	Verified	‘src/scripts/analysis
C-195	Norm submultiplicativity:——ab——¡= k*——a——*——b——with k = 1.0 exactly at composition dims (¡=8). Beyond, k decreases with...	Verified	‘src/scripts/analysis
C-196	Artin’s theorem: the subalgebra generated by any two elements {a, b, ab, ba, ...} is associative at dim ¡= 8...	Verified	‘src/scripts/analysis
C-197	Associator norm scaling: mean——A(a,b,c)——= 0 at dim¡=4 (associative), ˜1.088 at dim=8, and converges to sqrt(2) ˜ 1.414...	Verified	‘src/scripts/analysis
C-198	Middle Moufang identity (ab)(ca) = a((bc)a) holds EXACTLY at dim ¡= 8 (alternative/Moufang). Fails at dim ¿= 16 with...	Verified	‘src/scripts/analysis
C-199	Left/right alternative laws a(ab) = (aa)b and (ba)a = b(aa) hold EXACTLY at dim ¡= 8. Both fail symmetrically at dim ¿=...	Verified	‘src/scripts/analysis
C-200	Commutator-associator Leibniz identity [a, bc] = [a,b]c + b[a,c] - A(a,b,c) + A(b,a,c) - A(b,c,a) holds EXACTLY at ALL...	Verified	‘src/scripts/analysis

ID	Statement	Status	Source
C-201	Iterated product norms:——aˆn——=——a——ˆn = 1.0 EXACTLY at ALL CD dimensions from 4 through 256, for all powers n = 2...	Verified	‘src/scripts/analysis
C-202	Nucleus dimension: at dim¡=4 (associative), Nuc(A) = entire algebra. At dim¿=8, Nuc(A) = R*e_0 (dimension 1). Only the...	Verified	‘src/scripts/analysis
C-203	Inner derivation D(a,b)(c) = A(a,b,c) - A(b,a,c) is generically NOT a derivation. The Leibniz rule D(xy) = D(x)y +...	Verified	‘src/scripts/analysis
C-204	Composition algebra test N(xy) = N(x)N(y): exact at dim¡=8 (Hurwitz theorem). Fails at dim¿=16 but mean ratio...	Verified	‘src/scripts/analysis
C-205	Associator kernel dimension: dim(ker T_{a,b}) = dim at dim¡=4 (all associative), exactly 4 at dim=8 (quaternionic...	Verified	‘src/scripts/analysis
C-206	Product commutativity defect——ab-ba——/——ab——= 0 at dim=2 (commutative). At dim¿=4: approaches 2.0 monotonically: 1.14...	Verified	‘src/scripts/analysis
C-207	Triple product norms——(ab)c——and——a(bc)——are both exactly 1.0 at dim¡=8 (composition). At dim¿=16, left and right...	Verified	‘src/scripts/analysis
C-208	Flexible nucleus is FULL at ALL CD dimensions (4-64 tested): every basis element e_i satisfies (x*e_i)*x = x*(e_i*x)...	Verified	‘src/scripts/analysis
C-209	Associator norm distribution: at dim=8, slightly platykurtic (kurt=-0.63, negatively skewed). At high dim, excess...	Verified	‘src/scripts/analysis
C-210	Jordan product {a,b} = (ab+ba)/2: the Jordan identity {a, {b, aˆ2}} = {{a,b}, aˆ2} holds EXACTLY at ALL CD dims (4-128...	Verified	‘src/scripts/analysis
C-211	Quadratic identity a(ba) = (ab)a holds EXACTLY at ALL CD dims (4-512) for arbitrary non-unit vectors. Max deviation =...	Verified	‘src/scripts/analysis
C-212	Cascade associator norms: A1 = A(a,b,c), A2 = A(A1,d,e), A3 = A(A2,f,g). Ratios——A2——/——A1——˜ 1.2-1.4 and——A3——/——A2——˜...	Verified	‘src/scripts/analysis
C-213	Left multiplication operator L_a (x -¿ ax) is isometric (all eigenvalue magnitudes = 1.0) at dim¡=8 (composition...	Verified	‘src/scripts/analysis
C-214	Right multiplication operator R_a (x -¿ xa) is isometric (all eigenvalue magnitudes = 1.0) at dim¡=8 (composition...	Verified	‘src/scripts/analysis
C-215	Third and fourth power associativity: (aˆ2)*a = a*(aˆ2) and (aˆ2)ˆ2 = a*(aˆ3) hold EXACTLY at ALL CD dims 4-512. Max...	Verified	‘src/scripts/analysis
C-216	Associator is fully alternating (skew-symmetric under all 6 permutations) at dim¡=8 (alternative algebras). At dim¿=16,...	Verified	‘src/scripts/analysis
C-217	Jordan product norm——{a,b}——for unit vectors: exactly 1.0 at dim=2 (commutative), then monotonically decreasing: 0.77...	Verified	‘src/scripts/analysis
C-218	Bilinear form B(a,b) = Re(a*conj(b)): the Gram matrix G[i,j] = B(e_i,e_j) equals the identity matrix EXACTLY at ALL CD...	Verified	‘src/scripts/analysis

ID	Statement	Status	Source
C-219	Trace formula: Tr(L_a) = Tr(R_a) = dim * Re(a) holds EXACTLY at ALL CD dims 4-128. The traces are exactly equal...	Verified	‘src/scripts/analysis
C-220	Norm product ratio——ab——/(——a——*——b——) = 1.0 EXACTLY at dim¡=8 (composition algebras). At dim¿=16, the ratio has mean...	Verified	‘src/scripts/analysis
C-221	Only trivial idempotents (0 and e_0) exist in CD algebras: e_0ˆ2 = e_0 and 0ˆ2 = 0 are EXACT at ALL dims 4-128. No...	Verified	‘src/scripts/analysis
C-222	Commutant dimension dim(C(a)) = {x : xa = ax} is EXACTLY 2 at ALL CD dims 4-64 for generic unit a. The commutant is...	Verified	‘src/scripts/analysis
C-223	Associator trilinear ratio——A(a,b,c)——/(——a——*——b——*——c——) = 0 at dim=4 (associative). Mean -¿ sqrt(2) ˜ 1.414 at high...	Verified	‘src/scripts/analysis
C-224	Inner derivation space dimension: D(a,b)(x) = A(a,b,x) - A(b,a,x) spans a 0-dim space at dim=4 (quaternions are...	Verified	‘src/scripts/analysis
C-225	Moufang identity (ma)(bm) = m((ab)m): satisfied by ALL basis elements at dim¡=8 (Moufang loop). At dim¿=16, exactly 2...	Verified	‘src/scripts/analysis
C-226	Power-associativity aˆm * aˆn = aˆ(m+n) holds EXACTLY at ALL CD dims 4-256 for all 10 pairs (m,n) with m+n ¡= 5. Max...	Verified	‘src/scripts/analysis
C-227	Subalgebra gen{a,b} dimension: 4 at dim=4 (full quaternion algebra), 4 at dim=8 (Artin’s theorem confirmed – 2...	Verified	‘src/scripts/analysis
C-228	Commutator norm ratio——[a,b]——/——ab——-¿ 2.0 as dim -¿ infinity. Values: 1.14 (dim=4), 1.60 (dim=8), 1.80 (dim=16), 1.90...	Verified	‘src/scripts/analysis
C-229	Frobenius inner product Tr(L_a * L_bˆT) = dim * ¡a,b¿ (Euclidean dot product) holds EXACTLY at ALL CD dims 4-64....	Verified	‘src/scripts/analysis
C-230	Re(ab) = Re(ba) holds EXACTLY at ALL CD dims 4-512. The real part of the product is symmetric even though the full...	Verified	‘src/scripts/analysis
C-231	Reverse associator: A(c,b,a) = -A(a,b,c) holds EXACTLY at ALL CD dims 8-256. The associator is antisymmetric under full...	Verified	‘src/scripts/analysis
C-232	Real part formula: Re(ab) = a_0*b_0 - sum_{k¿=1} a_k*b_k (Lorentzian inner product) holds EXACTLY at ALL CD dims 2-512....	Verified	‘src/scripts/analysis
C-233	Left alternative law x(xy) = (xx)y holds EXACTLY at dim¡=8, fails at dim¿=16. Mean failure grows from 0.51 (dim=16) to...	Verified	‘src/scripts/analysis
C-234	Jordan norm scaling:——{a,b}——˜ C/sqrt(dim) with C ˜ 1.57. Power-law fit gives slope = -0.502 (expect -0.5)....	Verified	‘src/scripts/analysis
C-235	Operator determinant: det(L_a) = +/-1 for unit a at dim¡=8 (L_a is orthogonal). At dim¿=16,—det(L_a)—collapses rapidly:...	Verified	‘src/scripts/analysis
C-236	Right Moufang identity (ab)(ca) = a((bc)a) holds EXACTLY at dim¡=8, fails at dim¿=16. Failure grows: mean 0.79 (dim=16)...	Verified	‘src/scripts/analysis

ID	Statement	Status	Source
C-237	Symmetrized associator: A(a,b,c)+A(b,a,c) = 0 and A(a,b,c)+A(a,c,b) = 0 at dim¡=8 (alternating property). Both fail...	Verified	‘src/scripts/analysis
C-238	Right alternative law y(xx) = (yx)x holds EXACTLY at dim¡=8, fails at dim¿=16. Mean failure grows from 0.51 (dim=16) to...	Verified	‘src/scripts/analysis
C-239	Norm power scaling:——aˆn——=——a——ˆn holds EXACTLY at ALL CD dims 4-256, for powers n=2,3,4,5. This is UNIVERSAL, not...	Verified	‘src/scripts/analysis
C-240	Adjoint map T_a(x) = a*x*conj(a) is an isometry (——T_a(x)——=——x——) EXACTLY at dim¡=8. At dim¿=16, T_a distorts norms:...	Verified	‘src/scripts/analysis
C-241	Pythagorean decomposition:——ab——ˆ2 =——S(a,b)——ˆ2 +——A(a,b)——ˆ2 where S=(ab+ba)/2 and A=(ab-ba)/2 holds EXACTLY at ALL...	Verified	‘src/scripts/analysis
C-242	Cayley-Dickson doubling formula (a,b)*(c,d) = (ac - conj(d)*b, d*a + b*conj(c)) is verified EXACTLY at all tested...	Verified	‘src/scripts/analysis
C-243	Quadratic form N(a) = a*conj(a): (1) N(a) is purely real at ALL CD dims (imaginary parts = 0). (2) Re(N(a)) =——a——ˆ2...	Verified	‘src/scripts/analysis
C-244	Inverse element: aˆ{-1} = conj(a)/——a——ˆ2 satisfies a*aˆ{-1} = aˆ{-1}*a = e_0 EXACTLY at ALL CD dims 2-256. Both left...	Verified	‘src/scripts/analysis
C-245	Artin’s theorem: the subalgebra generated by any 2 elements is associative at dim¡=8 (alternative algebras). Verified...	Verified	‘src/scripts/analysis
C-246	Nucleus N(A) = {n : A(n,x,y) = A(x,n,y) = A(x,y,n) = 0 for all x,y}. At dim=4 (quaternions, associative): nucleus =...	Verified	‘src/scripts/analysis
C-247	Jacobi defect: [a,[b,c]]+[b,[c,a]]+[c,[a,b]] = 0 at dim¡=4 (associative: commutator forms Lie algebra). At dim=8...	Verified	‘src/scripts/analysis
C-248	Norm decomposition: 4*——ab——ˆ2 =——{a,b}——ˆ2 +——[a,b]——ˆ2 holds EXACTLY at ALL CD dims 2-512 (equivalent to C-241...	Verified	‘src/scripts/analysis
C-249	Trace of right multiplication product: Tr(R_a R_bˆT) = dim*¡a,b¿ holds EXACTLY at ALL CD dims 4-64. This mirrors C-229...	Verified	‘src/scripts/analysis
C-250	Left Bol identity ((ab)c)b = a((bc)b) holds EXACTLY at dim¡=8, fails at dim¿=16. Mean failure grows from 0.79 (dim=16)...	Verified	‘src/scripts/analysis
C-251	Eigenvalue spectrum of L_a: all eigenvalues lie on the unit circle (—lambda—=1) at dim¡=8 (L_a orthogonal). At dim¿=16,...	Verified	‘src/scripts/analysis
C-252	Commutator algebra dimension: span{[e_i, e_j]} has rank 0 at dim=2 (commutative), and rank dim-1 at dim¿=4. The...	Verified	‘src/scripts/analysis
C-253	Associator norm scaling:——A(a,b,c)——for unit vectors converges to sqrt(2) ˜ 1.4142 as dim -¿ infinity. Zero at dim=4...	Verified	‘src/scripts/analysis
C-254	Conjugate reversal: conj(ab) = conj(b)*conj(a) holds EXACTLY at ALL CD dims 2-256. Conjugation is a universal...	Verified	‘src/scripts/analysis

ID	Statement	Status	Source
C-255	Left multiplication operator: L_a^2 = L_{a^2} (as matrices) holds EXACTLY at dim¡=8 and fails at dim¿=16. This is...	Verified	‘src/scripts/analysis
C-256	R_a and L_a eigenvalue spectra match (sorted—eigenvalues—identical) at ALL CD dims 4-64. This verifies C-214 (spectral...	Verified	‘src/scripts/analysis
C-257	Associator norm concentration: CV(——A——) = std/mean decreases monotonically from 0.30 (dim=8) to 0.05 (dim=512). CV...	Verified	‘src/scripts/analysis
C-258	Product norm ratio:——ab——/(——a——*——b——) = 1.0 EXACTLY at dim¡=8 (composition property). At dim¿=16, the ratio has mean...	Verified	‘src/scripts/analysis
C-259	Doubling-level associator: at dim=16, left-half elements (octonion subalgebra) have——A——˜ 1.12 (matching octonion-level...	Verified	‘src/scripts/analysis
C-260	Associator 4-form: ¡A(a,b,c), d¿ is alternating (antisymmetric under adjacent swaps) at dim¡=8. Both swap(1,2) and...	Verified	‘src/scripts/analysis
C-261	Center Z(A) = full algebra (dim 2) at dim=2 (complex, commutative). Center = 1 (scalars R*e_0) at dim¿=4. This verifies...	Verified	‘src/scripts/analysis
C-262	Associator trilinearity: A(a+b,c,d) = A(a,c,d) + A(b,c,d) and A(alpha*a,c,d) = alpha*A(a,c,d), likewise in 2nd and 3rd...	Verified	‘src/scripts/analysis
C-263	Product of inverses: (ab)^{-1} = b^{-1}*a^{-1} holds EXACTLY at dim¡=8 (composition algebras). Fails at dim¿=16 with...	Verified	‘src/scripts/analysis
C-264	Commutator-to-associator norm ratio:——[a,b]——/——A(a,b,c)——-¿ sqrt(2) ˜ 1.414 as dim -¿ infinity. This follows...	Verified	‘src/scripts/analysis
C-265	Inner derivation D(a,b)(x) = [[a,b],x] - 3*A(a,b,x) satisfies the Leibniz rule D(xy) = D(x)y + xD(y) at dim¡=8...	Verified	‘src/scripts/analysis
C-266	Flexible nucleus = full algebra at ALL CD dims 4-128. Every element satisfies (xa)x = x(ax), confirming universal...	Verified	‘src/scripts/analysis
C-267	Moufang identity a(b(ac)) = ((ab)a)c holds EXACTLY at dim¡=8 (Moufang loop). Fails at dim¿=16 with violations growing...	Verified	‘src/scripts/analysis
C-268	Quadratic identity: x^2 - 2*Re(x)*x +——x——^2*e_0 = 0 holds EXACTLY at ALL CD dims 2-256. Every CD element satisfies a...	Verified	‘src/scripts/analysis
C-269	Power-norm for non-unit vectors:——x^n——=——x——^n at ALL CD dims 4-128 for n=2,3,4,5, with non-unit vectors of varying...	Verified	‘src/scripts/analysis
C-270	Di-associator (ax)b - a(xb) = 0 at dim¡=4 (associative). Nonzero at dim¿=8 with mean norm approaching sqrt(2) ˜ 1.414...	Verified	‘src/scripts/analysis
C-271	Artin’s theorem: subalgebra generated by any two elements is associative at dim¡=8 (alternative algebras). All products...	Verified	‘src/scripts/analysis
C-272	Trace of commutator: Tr(L_{[a,b]}) = 0 at ALL CD dims 4-128. This follows from Tr(L_a) = dim*Re(a) (C-230) and...	Verified	‘src/scripts/analysis

ID	Statement	Status	Source
C-273	Nucleus is NOT an ideal at dim¿=8. For n = lambda*e_0 (scalar, in nucleus) and arbitrary a, the product na = lambda*a...	Verified	‘src/scripts/analysis
C-274	Polarization identity: both standard (4¡a,b¿ =——a+b——ˆ2 -——a-b——ˆ2) and CD-specific (Re(conj(a)*b) = ¡a,b¿) hold...	Verified	‘src/scripts/analysis
C-275	Jordan product power-associativity: (a.b).(a.a) = a.(b.(a.a)) where a.b = (ab+ba)/2 holds at ALL CD dims 4-128. The...	Verified	‘src/scripts/analysis
C-276	Jordan triple product {a,b,c} = a.(b.c) + c.(b.a) - b.(a.c) is symmetric in (a,c): {a,b,c} = {c,b,a} at ALL CD dims...	Verified	‘src/scripts/analysis
C-277	Basis element squares: e_kˆ2 = -e_0 for all k¿=1 at ALL CD dims 2-128. e_0ˆ2 = +e_0 (identity). All dim-1 imaginary...	Verified	‘src/scripts/analysis
C-278	No nilpotent elements: xˆn != 0 for any nonzero x at ALL CD dims 8-128. This follows from——xˆn——=——x——ˆn (C-239/C-269)...	Verified	‘src/scripts/analysis
C-279	Alternative nucleus: ALL random elements satisfy both left and right alternative laws at dim¡=8. NO random element...	Verified	‘src/scripts/analysis
C-280	L_a eigenvalue distribution: at dim¡=8, all eigenvalues of L_a (unit a) lie exactly on the unit circle (—lambda—=1,...	Verified	‘src/scripts/analysis
C-281	Near-zero-divisor product norm: min(——ab——/(——a——*——b——)) = 1.0 exactly at dim¡=8. At dim¿=16, min ratio drops below...	Verified	‘src/scripts/analysis
C-282	Subalgebra embedding chain: R c C c H c O c S c P verified within dim=64. Elements with support in the first d...	Verified	‘src/scripts/analysis
C-283	Associator mean norm convergence: mean(——A(a,b,c)——) for unit vectors increases monotonically from 1.12 (dim=8) to 1.41...	Verified	‘src/scripts/analysis
C-284	Anti-commutator norm ratio:——{a,b}——/(2*——a——*——b——) = 1.0 at dim=2 (commutative), decreasing to ˜0.10 at dim=256. The...	Verified	‘src/scripts/analysis
C-285	Power tower:——xˆ{2ˆk}——=——x——ˆ{2ˆk} remains exact to machine precision through k=8 (xˆ{256}) at ALL CD dims 4-64 for...	Verified	‘src/scripts/analysis
C-286	Norm product identity: a*conj(a) = conj(a)*a = ——a——ˆ2 * e_0 holds EXACTLY at ALL CD dims 2-256. Both orderings produce...	Verified	‘src/scripts/analysis
C-287	Imaginary product structure: for pure imaginary a,b (Re=0), Re(ab) = -¡a,b¿ (negative inner product) holds EXACTLY at...	Verified	‘src/scripts/analysis
C-288	L_a eigenvalue conjugate pairing: all eigenvalues of the left-multiplication matrix L_a come in conjugate pairs (real...	Verified	‘src/scripts/analysis
C-289	n-fold product norm: ——a1*a2*...*an——/ prod(——ai——) = 1.0 exactly at dim¡=8 for all n=3,5,8 (composition algebra). At...	Verified	‘src/scripts/analysis
C-290	Associator antisymmetry: A(a,b,c) = -A(b,a,c) holds exactly at dim¡=8 (alternating associator, swap ratio = 0). At...	Verified	‘src/scripts/analysis

ID	Statement	Status	Source
C-291	Generated subalgebra dimension: every nonzero element x of a CD algebra generates a 2-dimensional subalgebra isomorphic...	Verified	‘src/scripts/analysis
C-292	Moufang identity a(b(ac)) = (a(ba))c holds EXACTLY at dim¡=8 (Moufang loop) and FAILS at dim¿=16. The failure ratio...	Verified	‘src/scripts/analysis
C-293	Conjugate anti-automorphism: conj(ab) = conj(b)*conj(a) holds at ALL CD dims 2-256 with EXACT zero error (bitwise...	Verified	‘src/scripts/analysis
C-294	Center of CD algebra: Z(A) = full algebra at dim=2 (C is commutative); Z(A) = R*e_0 (scalars only) at dim¿=4. Verified...	Verified	‘src/scripts/analysis
C-295	Cyclic associator sum: A(a,b,c) + A(b,c,a) + A(c,a,b) = 0 at dim¡=4 (associative), = 3*A(a,b,c) EXACTLY at dim=8...	Verified	‘src/scripts/analysis
C-296	Left-right multiplication intertwining: xa = conj(conj(a)*conj(x)) holds at ALL CD dims 2-256 with EXACT zero error...	Verified	‘src/scripts/analysis
C-297	Product of conjugates: conj(a)*conj(b) = conj(ba) holds at ALL CD dims 2-256 with EXACT zero error (bitwise identical)....	Verified	‘src/scripts/analysis
C-298	Trace of left-multiplication: Tr(L_a) = dim * Re(a) holds EXACTLY at ALL CD dims 4-64. This is a universal spectral...	Verified	‘src/scripts/analysis
C-299	Right Bol identity (ab)(ca) = a((bc)a) holds EXACTLY at dim¡=8 and FAILS at dim¿=16. The failure ratio increases...	Verified	‘src/scripts/analysis
C-300	Commutator-anticommutator decomposition: ——[a,b]——ˆ2 + ——{a,b}——ˆ2 = 4*——ab——ˆ2 holds EXACTLY at ALL CD dims 2-256....	Verified	‘src/scripts/analysis
C-301	Derivation algebra dimension: dim(Der(R))=0, dim(Der(C))=0, dim(Der(H))=3 (=so(3)), dim(Der(O))=14 (=G2),...	Verified	‘src/scripts/analysis
C-302	Associator (1,3)-swap antisymmetry: A(a,b,c) = -A(c,b,a) holds at ALL CD dims (universal). This is stronger than...	Verified	‘src/scripts/analysis
C-303	Real part symmetry: Re(ab) = Re(ba) holds at ALL CD dims 2-256 (universal). This follows from Re(ab) = ¡a, conj(b)¿ and...	Verified	‘src/scripts/analysis
C-304	Inverse element: a * conj(a)/——a——ˆ2 = conj(a)/——a——ˆ2 * a = e_0 holds EXACTLY at ALL CD dims 2-256. Every nonzero...	Verified	‘src/scripts/analysis
C-305	L_a and R_a spectral equivalence: the multisets of eigenvalue magnitudes of L_a and R_a are identical at ALL CD dims...	Verified	‘src/scripts/analysis
C-306	Flexible product norm: ——a(ba)——= ——a——ˆ2*——b——exactly at dim¡=8 (composition). At dim¿=16, the ratio...	Verified	‘src/scripts/analysis
C-307	Scalar triple product associativity: Re(a(bc)) = Re((ab)c) holds EXACTLY at ALL CD dims 4-256. The real part of a...	Verified	‘src/scripts/analysis
C-308	Inverse composition: (ab)ˆ{-1} = bˆ{-1}*aˆ{-1} holds EXACTLY at dim¡=8 and FAILS at dim¿=16. The failure ratio is 0.14...	Verified	‘src/scripts/analysis

ID	Statement	Status	Source
C-309	Commutator norm scaling: ——[a,b]——for unit vectors = 0 at dim=2 (commutative), increases through 1.14 (dim=4), 1.60...	Verified	‘src/scripts/analysis
C-310	CD doubling construction: (a,b)*(c,d) = (ac - conj(d)*b, d*a + b*conj(c)) is verified EXACTLY at all levels dim=4...	Verified	‘src/scripts/analysis
C-311	Associator (2,3)-swap: A(a,b,c) = -A(a,c,b) holds EXACTLY at dim¡=8 and FAILS at dim¿=16. The swap ratio increases:...	Verified	‘src/scripts/analysis
C-312	Chebyshev relation: for unit x, Re(xˆn) = T_n(Re(x)) where T_n is the n-th Chebyshev polynomial. Verified for...	Verified	‘src/scripts/analysis
C-313	Inner product adjoint: ¡ab, c¿ = ¡b, conj(a)*c¿ holds EXACTLY at ALL CD dims 4-256 (universal). This means L_aˆ* =...	Verified	‘src/scripts/analysis
C-314	Norm of sum: ——a+b——ˆ2 = ——a——ˆ2 + ——b——ˆ2 + 2*Re(conj(a)*b) holds EXACTLY at ALL CD dims 2-256. This is the standard...	Verified	‘src/scripts/analysis
C-315	Polarization identity: 4*Re(conj(a)*b) = ——a+b——ˆ2 - ——a-b——ˆ2 holds EXACTLY at ALL CD dims 2-256. This recovers the...	Verified	‘src/scripts/analysis
C-316	L_a quadratic identity: L_aˆ2 - 2*Re(a)*L_a + ——a——ˆ2*I = 0 holds at dim¡=8 (composition algebras) and FAILS at...	Verified	‘src/scripts/analysis
C-317	Product norm ratio variance: Var(——ab——/(——a————b——)) = 0 exactly at dim¡=8 (norm multiplicativity) and ¿0 at dim¿=16....	Verified	‘src/scripts/analysis
C-318	Associator norm concentration: CV of ——A(a,b,c)——scales as dimˆalpha with alpha = -0.448 (near the -0.5 expected from...	Verified	‘src/scripts/analysis
C-319	Flexibility: (ab)a = a(ba) holds EXACTLY at ALL CD dims 2-256. This is a defining property of flexible algebras and...	Verified	‘src/scripts/analysis
C-320	Jordan identity: (aˆ2*b)*a = aˆ2*(b*a) holds EXACTLY at ALL CD dims 4-128. This is the right Jordan identity, which...	Verified	‘src/scripts/analysis
C-321	Minimal polynomial: every CD element x satisfies xˆ2 - 2*Re(x)*x + ——x——ˆ2*e_0 = 0 at ALL dims 4-128. This means every...	Verified	‘src/scripts/analysis
C-322	Left-alternative identity: (aa)b = a(ab) holds EXACTLY at dim¡=8 and FAILS at dim¿=16. Diffs grow: 65.6 (dim=16), 119.3...	Verified	‘src/scripts/analysis
C-323	Right-alternative identity: (ba)a = b(aa) holds EXACTLY at dim¡=8 and FAILS at dim¿=16. This mirrors the...	Verified	‘src/scripts/analysis
C-324	Commutator energy partition: ——[a,b]——ˆ2/——ab——ˆ2 increases from 0.0 (dim=2, commutative) through 1.51 (dim=4), 2.67...	Verified	‘src/scripts/analysis
C-325	Nucleus structure: e_0 is in the nucleus (left, middle, and right) at ALL CD dims. At dim¡=4 (associative): the nucleus...	Verified	‘src/scripts/analysis
C-326	Norm submultiplicativity: ——ab——¡= C(dim)*——a——*——b——with C(dim)=1.0 exactly at dim¡=8 (norm multiplicativity) and...	Verified	‘src/scripts/analysis

ID	Statement	Status	Source
C-327	Left Moufang identity: a(b(ac)) = ((ab)a)c holds EXACTLY at dim¡=8 and FAILS at dim¿=16. Moufang identities are...	Verified	‘src/scripts/analysis
C-328	Power-associativity: aˆ2*aˆ3 = aˆ5 and aˆ3*aˆ3 = aˆ6 hold at ALL CD dims 4-256. This verifies that CD algebras are...	Verified	‘src/scripts/analysis
C-329	L_a eigenvalue structure: at dim¡=8 (composition algebras), L_a has exactly 1 distinct eigenvalue magnitude =...	Verified	‘src/scripts/analysis
C-330	Fourth-power norm: ——aˆ2——ˆ2 = ——a——ˆ4 at ALL CD dims 2-256. This follows from the quadratic identity (C-321): aˆ2 =...	Verified	‘src/scripts/analysis
C-331	Bimodule commutation: L_a*R_b = R_b*L_a (i.e. a(xb) = (ax)b for all x) holds iff dim¡=4 (associative). At dim¿=8, L and...	Verified	‘src/scripts/analysis
C-332	Artin’s theorem: (a, b, ab) = 0 (the associator of a, b, and their product vanishes) at dim¡=8 (alternative algebras)....	Verified	‘src/scripts/analysis
C-333	Conjugate product norm: ——conj(a)*a——= ——a——ˆ2 at ALL CD dims 2-256. Since conj(a)*a = ——a——ˆ2*e_0 exactly (C-286), the...	Verified	‘src/scripts/analysis
C-334	Exponential map: for purely imaginary unit u (Re(u)=0, ——u——=1), exp(t*u) = cos(t)*e_0 + sin(t)*u. Verified via 20-term...	Verified	‘src/scripts/analysis
C-335	Squaring Lipschitz constant: ——aˆ2 - bˆ2——¡= C*——a-b——*(——a——+——b——) with C¡=1 at dim¡=8 and C decreasing with dim:...	Verified	‘src/scripts/analysis
C-336	Trace of left multiplication: Tr(L_a) = dim*Re(a) at ALL CD dims 2-128. Each diagonal entry L_a[j,j] = Re(a) because...	Verified	‘src/scripts/analysis
C-337	Inner derivation: D(a,b) = [L_a, L_b] - L_{[a,b]} = 0 at dim¡=4 (associative) and D != 0 at dim¿=8. At dim=8, D(a,b)...	Verified	‘src/scripts/analysis
C-338	Pythagorean decomposition: ——a——ˆ2 = Re(a)ˆ2 + ——Im(a)——ˆ2 at ALL CD dims 2-256. This is trivially the Euclidean norm...	Verified	‘src/scripts/analysis
C-339	Imaginary square: for purely imaginary u (Re(u)=0), uˆ2 = -——u——ˆ2*e_0 at ALL CD dims 2-256. This follows from the...	Verified	‘src/scripts/analysis
C-340	Norm of product of conjugates: ——conj(a)*conj(b)——= ——a——*——b——iff dim¡=8 (norm multiplicativity via conjugates). Since...	Verified	‘src/scripts/analysis
C-341	Right norm identity: Re(a*conj(a)) = ——a——ˆ2 at ALL CD dims 2-256. Since a*conj(a) = ——a——ˆ2*e_0 (both orderings give...	Verified	‘src/scripts/analysis
C-342	Real part of square: Re(aˆ2) = 2*Re(a)ˆ2 - ——a——ˆ2 at ALL CD dims 2-256. This is the Re-component of the quadratic...	Verified	‘src/scripts/analysis
C-343	Determinant of L_a: —det(L_a)—= ——a——ˆdim iff dim¡=8 (composition algebras). At dim¡=8, L_a is a scaled orthogonal...	Verified	‘src/scripts/analysis
C-344	Trilinear associativity: Re((ab)c) = Re(a(bc)) at ALL CD dims 2-256. The scalar triple product is associative even when...	Verified	‘src/scripts/analysis

ID	Statement	Status	Source
C-345	Cyclic trilinear form: Re(a(bc)) = Re(b(ca)) = Re(c(ab)) at ALL CD dims 2-256. The scalar triple product is invariant...	Verified	‘src/scripts/analysis
C-346	Power norm: ——aˆn——= ——a——ˆn for n=2,3,4,5 at ALL CD dims 4-128. Follows from the quadratic identity aˆ2 = 2*Re(a)*a -...	Verified	‘src/scripts/analysis
C-347	Commutator Jacobi identity: [[a,b],c] + [[b,c],a] + [[c,a],b] = 0 iff dim¡=4 (associative algebras). At dim¿=8, Jacobi...	Verified	‘src/scripts/analysis
C-348	Middle Moufang identity: (ab)(ca) = a((bc)a) at ALL CD dims¡=8 (alternative algebras). Fails at dim¿=16 with diffs...	Verified	‘src/scripts/analysis
C-349	Imaginary product norm Pythagorean: ——Im(ab)——ˆ2 = ——ab——ˆ2 - Re(ab)ˆ2 at ALL CD dims 2-256. This is the Pythagorean...	Verified	‘src/scripts/analysis
C-350	Left-multiplication square identity: L_{aˆ2} = (L_a)ˆ2 iff dim¡=8 (alternative algebras). This is the left-alternative...	Verified	‘src/scripts/analysis
C-351	Right Moufang identity: ((ab)c)b = a(b(cb)) at ALL CD dims¡=8 (alternative algebras). Fails at dim¿=16 with diffs...	Verified	‘src/scripts/analysis
C-352	Malcev identity: J(x,y,xz) = J(x,y,z)*x where J is the Jacobian of double commutators. Holds at dim¡=8 (octonions form...	Verified	‘src/scripts/analysis
C-353	Schafer identity: J(a,b,c) = 6*[a,b,c] in alternative algebras (dim¡=8). The Jacobian of double commutators equals 6...	Verified	‘src/scripts/analysis
C-354	Commutator norm scaling: ——[a,b]——ˆ2 / (——a——ˆ2 * ——b——ˆ2) converges to 4 as dim -¿ infinity. Mean ratio: 1.50 (dim=4),...	Verified	‘src/scripts/analysis
C-355	Quadratic identity: xˆ2 - 2*Re(x)*x + ——x——ˆ2*e_0 = 0 at ALL CD dims 2-256. Every element of a CD algebra satisfies a...	Verified	‘src/scripts/analysis
C-356	Artin’s theorem (2-generated subalgebra associativity): any subalgebra generated by two elements is associative at...	Verified	‘src/scripts/analysis
C-357	Conjugation anti-automorphism: conj(ab) = conj(b)*conj(a) at ALL CD dims 2-256. This is EXACTLY zero (not just...	Verified	‘src/scripts/analysis
C-358	Associator norm scaling: ——[a,b,c]——ˆ2 / (——a——ˆ2*——b——ˆ2*——c——ˆ2) is zero at dim=4 (associative), nonzero at dim¿=8,...	Verified	‘src/scripts/analysis
C-359	Adjoint identity: L_aˆT = L_{conj(a)} at ALL CD dims 2-64. The transpose of the left-multiplication matrix equals left...	Verified	‘src/scripts/analysis
C-360	Eigenvalue pattern of L_u for pure imaginary unit u: L_u is skew-symmetric at ALL dims (all eigenvalues purely...	Verified	‘src/scripts/analysis
C-361	Trace form symmetry: Re(ab) = Re(ba) at ALL CD dims 2-256. The real part of a product is commutative. This follows from...	Verified	‘src/scripts/analysis
C-362	Two-sided inverse: a * (conj(a)/——a——ˆ2) = (conj(a)/——a——ˆ2) * a = e_0 at ALL CD dims 2-256. Every nonzero CD element...	Verified	‘src/scripts/analysis

ID	Statement	Status	Source
C-363	Polarization identity: ——a+b——ˆ2 = ——a——ˆ2 + ——b——ˆ2 + 2*Re(conj(a)*b) at ALL CD dims 2-256. This is the standard inner...	Verified	‘src/scripts/analysis
C-364	Inverse multiplicativity: (ab)ˆ{-1} = bˆ{-1}*aˆ{-1} iff dim¡=8 (alternative algebras). At dim¿=16, the relation fails...	Verified	‘src/scripts/analysis
C-365	Left Bol identity: a(b(ac)) = (a(ba))c iff dim¡=8. In a Moufang loop, left and right Bol identities both hold. At...	Verified	‘src/scripts/analysis
C-366	Left-right operator commutation: L_a*R_b = R_b*L_a iff dim¡=4 (associative). At dim¿=8, L_a and R_b generically do not...	Verified	‘src/scripts/analysis
C-367	Cayley-Dickson doubling construction: (a,b)*(c,d) = (ac - conj(d)*b, d*a + b*conj(c)) verified at all dims 4-128...	Verified	‘src/scripts/analysis
C-368	Center of CD algebra: at dim=2 (complex), center = full algebra (commutative). At dim¿=4, center = R*e_0 (scalars...	Verified	‘src/scripts/analysis
C-369	Derivation algebra dimensions: Der(C) = 0 (dim=2), Der(H) = so(3) dim 3 (dim=4), Der(O) = g2 dim 14 (dim=8). Computed...	Verified	‘src/scripts/analysis
C-370	Right Bol identity: ((xy)z)y = x(y(zy)) iff dim¡=8 (alternative algebras). At dim¿=16, fails with diffs ˜320-10800....	Verified	‘src/scripts/analysis
C-371	Automorphism group dimension: Aut(C)=0 (dim=2), Aut(H)=3=SO(3) (dim=4), Aut(O)=14=G2 (dim=8). Computed via null space...	Verified	‘src/scripts/analysis
C-372	N-fold associator growth: at dim¡=4 (associative algebras), all n-fold nested associators vanish exactly. At dim¿=8,...	Verified	‘src/scripts/analysis
C-373	Nucleus structure: Nuc(A) = full algebra at dim¡=4 (associative). At dim¿=8, Nuc(A) = R*e_0 (scalars only). Scalar...	Verified	‘src/scripts/analysis
C-374	Normed division algebra (Hurwitz theorem):——xy——=——x——*——y——exactly iff dim in {1,2,4,8}. At dim¿=16, norm...	Verified	‘src/scripts/analysis
C-375	Anti-involution properties of CD conjugation: (1) conj(conj(a)) = a (involutive),...	Verified	‘src/scripts/analysis
C-376	Flexibility identity: a(ba) = (ab)a holds universally at all CD dimensions 4-128. This follows from the quadratic...	Verified	‘src/scripts/analysis
C-377	Power-associativity: xˆm * xˆn = xˆ{m+n} for m,n in {2,3} holds universally at all CD dimensions 4-128 with unit-norm...	Verified	‘src/scripts/analysis
C-378	Inner product composition identity: ¡xy,xz¿ =——x——ˆ2 ¡y,z¿ holds iff dim¡=8 (composition algebras). This is the...	Verified	‘src/scripts/analysis
C-379	Frobenius norm of L_a:——L_a——_Fˆ2 = dim *——a——ˆ2 holds universally at all CD dimensions 2-64. This follows from...	Verified	‘src/scripts/analysis
C-380	Left regular representation defect:——L_{ab} - L_a*L_b——_F / (——a——*——b——*dim) is zero at dim¡=4 (associative) and...	Verified	‘src/scripts/analysis

ID	Statement	Status	Source
C-381	Anticommutator norm scaling:——{a,b}——ˆ2/(——a——ˆ2*——b——ˆ2) = 4.0 exactly at dim=2 (commutative, composition). Decreases...	Verified	‘src/scripts/analysis
C-382	Right alternative identity: (ab)b = a(bb) holds iff dim¡=8 (alternative algebras). At dim¿=16, absolute deviations...	Verified	‘src/scripts/analysis
C-383	Left alternative identity: a(ab) = (aa)b holds iff dim¡=8 (alternative algebras). At dim¿=16, absolute deviations...	Verified	‘src/scripts/analysis
C-384	Jordan identity: aˆ2(ba) = (aˆ2 b)a holds universally at all CD dimensions 4-128. This follows from flexibility...	Verified	‘src/scripts/analysis
C-385	Commutator Jacobi identity: J(a,b,c) = [a,[b,c]] + [b,[c,a]] + [c,[a,b]] = 0 iff dim¡=4 (associative). At dim¿=8, the...	Verified	‘src/scripts/analysis
C-386	Trace bilinear form: Re(x * conj(y)) = dot(x, y) universally at all CD dimensions 2-128. The trace form T(x,y) =...	Verified	‘src/scripts/analysis
C-387	Norm product ratio:——ab——ˆ2/(——a——ˆ2*——b——ˆ2) = 1 exactly iff dim¡=8. At dim¿=16, the ratio has mean ˜1.0 but nonzero...	Verified	‘src/scripts/analysis
C-388	Third Moufang relative error: (ab)(ca) = a((bc)a) holds exactly iff dim¡=8. At dim¿=16, mean relative error grows from...	Verified	‘src/scripts/analysis
C-389	Right regular representation defect: R_{ab} = R_b*R_a iff dim¡=4 (associative). At dim¿=8, normalized defect ˜0.40...	Verified	‘src/scripts/analysis
C-390	Commutativity measure:——ab-ba——/——ab——= 0 at dim=2, increases monotonically, converges to 2.0 as dim-¿inf. At dim=256,...	Verified	‘src/scripts/analysis
C-391	Associator norm for unit elements:——[a,b,c]——= 0 at dim¡=4, mean ˜1.07 at dim=8, converging to ˜1.40 at high dim. The...	Verified	‘src/scripts/analysis
C-392	Multiplication table structure: for all CD algebras dim 2-64, every basis product e_i*e_j is exactly +/-e_k for some...	Verified	‘src/scripts/analysis
C-393	Eigenvalue distribution of L_u (pure imaginary unit): at dim¡=8, all nonzero eigenvalue magnitudes = 1.0 exactly (L_u...	Verified	‘src/scripts/analysis
C-394	Determinant of L_a: det(L_a) =——a——ˆdim exactly iff dim¡=8 (composition algebras). At dim¿=16, the...	Verified	‘src/scripts/analysis
C-395	Adjoint identity: L_aˆT = L_{conj(a)} holds universally at all CD dimensions 2-64. Exactly zero difference at all dims....	Verified	‘src/scripts/analysis
C-396	Gram matrix spectrum: L_a*L_aˆT =——a——ˆ2*I iff dim¡=8. At dim¿=16, normalized eigenvalues spread: n_distinct = 3...	Verified	‘src/scripts/analysis
C-397	Zero divisor existence: no zero divisors among diagonal 2-blades at dim¡=8. At dim¿=16, zero divisors exist (found by...	Verified	‘src/scripts/analysis
C-398	Product of conjugates: conj(b)*conj(a) = conj(a*b) holds universally at all CD dimensions 2-256. Exactly zero...	Verified	‘src/scripts/analysis

ID	Statement	Status	Source
C-399	Idempotent structure: the ONLY idempotents in any CD algebra are 0 and e_0 (the identity). This follows from the...	Verified	‘src/scripts/analysis
C-400	Metamaterials can emulate Alcubierre warp drive metrics for electromagnetic waves (Analog Gravity).	Verified	‘docs/external_sourc
C-401	A Casimir cavity (1um sphere in 4um cylinder) generates the negative energy density required for a nanoscale warp...	Theoretical	‘docs/external_sourc
C-402	Metamaterial Gravitational Coupling can reduce warp drive energy requirements to achievable levels.	Refuted	‘docs/external_sourc
C-403	Geometry from spectral data must be stated at spectral-triple strength (A,H,D_geom) or equivalent; eigenvalues alone...	Verified	‘docs/convos/CONV
C-404	Bulk locality can be organized by boundary modular data K_A=-log rho_A and entanglement wedge reconstruction;...	Verified	‘docs/convos/CONV
C-405	Observers can be modeled as (approximate) correctable record algebras selected by open-system dynamics L_open...	Verified	‘docs/convos/CONV
C-406	TSCP/box-kite sky mapping must be invariant under the relevant algebraic symmetries (or else explicitly enumerate...	Verified	‘docs/convos/CONV
C-407	Reported sky-alignment p-values must include explicit trial-factor accounting (look-elsewhere) across tuned degrees of...	Verified	‘docs/convos/CONV
C-408	Every thesis-level hypothesis must have symmetric falsification boundaries: a disconfirmation threshold (N_min,...	Verified	‘docs/convos/CONV
C-409	”Interleaved I-Beam” spaceplate design targets high refractive index via capacitive loading (metal/dielectric stack).	Verified	‘crates/materials_co
C-410	One-loop photon-graviton mixing in constant EM fields includes non-zero tadpole contributions (Ahmadiniaz et al. 2026),...	Verified	‘docs/BIBLIOGRAI
C-411	Spontaneous four-wave mixing (SFWM) dominates in thin LN layers due to relaxed phase matching, enabling flat entangled...	Verified	‘docs/BIBLIOGRAI
C-412	”Holographic Entropy Trap” 5-phase mechanism (Injection -¿ Lensing -¿ Sedenion Resonance -¿ Parton Decay -¿ Extraction)...	Verified	‘data/artifacts/imag
C-413	Determinant of left-multiplication operator ‘det(L_a)‘ equals ——a——ˆdim exactly iff dim ¡= 8; collapses to near-zero at...	Verified	‘src/scripts/analysis
C-414	Adjoint identity ‘L_aˆT = L_{conj(a)}‘ holds universally at all Cayley-Dickson dimensions (tested up to 64).	Verified	‘src/scripts/analysis
C-415	Gram spectrum ‘L_a L_aˆT‘ is a scalar multiple of identity ——a——ˆ2 I iff dim ¡= 8; eigenvalues spread significantly at...	Verified	‘src/scripts/analysis
C-416	The only idempotent elements in any Cayley-Dickson algebra are 0 and 1 (the identity).	Verified	‘src/scripts/analysis
C-417	Hypothesis: Ray capture efficiency in fractal metamaterials correlates with Sedenion Zero Divisor density; ”Holographic...	Closed/Research-Program	‘docs/external_sourc

ID	Statement	Status	Source
C-418	Material Database tracks temperature-dependent phase transitions (e.g., Ice Ih/VII/X) and wide-spectrum optical...	Verified	‘crates/materials_co
C-419	Digital Matter BOM generation pipeline produces fabrication-ready CSVs tracking layer stoichiometry, mass density...	Verified	‘src/scripts/engineer
C-420	Automated CAD generation outputs OpenSCAD geometry and SVG lithography masks for metamaterial nanostructures, linking...	Verified	‘src/scripts/engineer
C-421	Metamaterial designs incorporate Rogers RT5880 carrier substrates and Gold/Silicon I-beam stacks to achieve...	Verified	‘crates/materials_co
C-422	Negative-dimension vacuum dynamics ($D \{}sim kˆ{-3}$) coupled with attractive self-interaction spontaneously generates...	Verified	‘src/scripts/simulati
C-423	Grand Unified Simulator v4 integrates CUDA-based relativistic ray tracing with robust FDFD electromagnetic field...	Verified	‘src/scripts/engineer
C-424	”Holographic Warp Gate” simulation harmonizes Alcubierre metric gradients with Kozyrev p-adic noise and Spaceplate...	Verified	‘src/scripts/engineer
C-425	Octonion-valued (8D) scalar field Hamiltonian with Stormer-Verlet symplectic integrator, 7 Noether charges from...	Verified	‘crates/algebra_core/
C-426	Pathion (32D) Zero-Divisor interaction matrix diagonalization provides a testbed for algebraic mass spectrum hypotheses...	Refuted	‘docs/external_sourc
C-427	Algebraic Metamaterial synthesis maps Cayley-Dickson structure constants to permittivity tensors and Clifford subspace...	Closed/Research-Program	‘docs/external_sourc
C-428	Integrated Warp Geodesic Simulation traces null rays through a Kerr metric background using ’NegDimCosmology’ expansion...	Verified	‘src/scripts/simulati
C-429	Kerr black hole shadow asymmetry due to frame dragging (a=0.99) validated: analytic Bardeen shadow boundary shows...	Verified	‘crates/gr_core/src/l
C-430	Negative Dimension Cosmology (NegDim) expansion history H(z) and luminosity distance D_L(z) deviate from standard...	Closed/Refuted	‘src/scripts/analysis
C-431	The Sedenion Zero Divisor manifold, when projected into 3D perturbation space (e2, e5, e14), forms a coherent,...	Verified	‘src/scripts/visualiza
C-432	Kerr geodesic trajectories validated against analytic Bardeen (1973) shadow boundary: Schwarzschild (a=0) shadow radius...	Verified	‘crates/gr_core/src/l
C-433	Hamiltonian Ray Equation solver (dT/ds = (grad_n - (T*grad_n)T)/n) implemented using 4th-order Runge-Kutta (RK4) for...	Verified	‘crates/optics_core/s
C-434	Material stack Silicon-Gold-Ice VIII reconciled with literature-supported optical constants (Si n=3.48, Au complex eps,...	Verified	‘docs/theory/WARP
C-435	”Power Pipeline” energy bookkeeping model (Photon intensity -¿ boundary absorption -¿ Plasmon field -¿ Parton jitter...	Verified	‘docs/theory/WARP
C-436	FRB comoving positions exhibit local ultrametric structure (C-071 follow-up, Direction 1).	Refuted	‘crates/gororoba_cli/

ID	Statement	Status	Source
C-437	Multi-attribute parameter spaces exhibit ultrametric structure under Euclidean distances in astrophysical catalogs with...	Verified	‘crates/gororoba_cli/
C-438	Repeating FRB temporal cascades exhibit ultrametric hierarchy (C-071 follow-up, Direction 3).	Verified	‘crates/gororoba_cli/
C-439	GW merger mass-ratio clustering exhibits ultrametric structure (C-071 follow-up, Direction 4).	Verified	‘crates/gororoba_cli/
C-440	Cross-dataset cosmic objects (FRBs+GW events) exhibit dendrogram ultrametricity in comoving 3D space (C-071 follow-up,...	Refuted	‘crates/gororoba_cli/
C-441	Bounce cosmology (quantum-corrected Friedmann equation) is disfavored by joint Pantheon+ SN Ia + DESI DR1 BAO fit;...	Verified	‘crates/cosmology_co
C-442	Multi-attribute Euclidean ultrametricity is specific to radio transient catalogs (FRB, pulsar) parameterized by DM+sky...	Refuted	‘crates/gororoba_cli/
C-443	Pathion (dim=32) zero-divisor complement graph decomposes into 15 connected components with two distinct motif types: 8...	Verified	‘crates/algebra_core/
C-444	CD motif components at dim=2^n correspond bijectively to points of PG(n-2,2) finite projective space. At dim=16 (7...	Verified	‘crates/algebra_core/
C-445	Motif class assignment within a given CD dimension is determined by a GF(2)-linear predicate: a weight vector w in...	Refuted	‘crates/algebra_core/
C-446	Sign-twist signature (4-bit encoding of sign products for cross-pair interactions) fully determines the zero-product...	Verified	‘crates/algebra_core/
C-447	Adaptive permutation testing (Besag-Clifford 1991) preserves type-I error rate at nominal alpha=0.05 while achieving...	Verified	‘crates/stats_core/sr
C-448	The Pathion Cubic Anomaly: the zero-divisor motif partition at dim=32 (8 heptacross + 7 mixed-degree) requires a...	Verified	‘crates/algebra_core/
C-449	Ultrametric Core Mining Hypothesis (UCMH): for multi-attribute astronomical catalogs, the minimal attribute subsets...	Verified	‘src/scripts/analysis
C-450	The external 64D pathion adjacency CSV (64x64 binary matrix, 64 nonzero entries) uses a basis-element-level node...	Inconclusive	‘crates/algebra_core/
C-451	128D Cayley-Dickson cross-validation: 4032 cross-pairs confirmed, XOR partner law universal (3968/3968 valid, mask=8),...	Verified	‘crates/algebra_core/
C-452	Cayley-Dickson basis elements at dim=256 embed into an 8-dimensional integer lattice with coordinates in {-1, 0, 1}....	Verified	‘crates/algebra_core/
C-453	The 8D lattice embedding extends to dims 512, 1024, and 2048 (all use 8D lattice, NOT log2(dim)-dimensional as...	Verified	‘crates/algebra_core/
C-454	De Marrais’s strut table from ”Flying Higher Than A Box-Kite” (unpublished, page 3) matches our computed strut tables...	Verified	‘crates/algebra_core/
C-455	Lattice-point differences between ZD-adjacent cross-assessor pairs at dim=16 (mapped to the 256D lattice) do NOT...	Refuted	‘crates/algebra_core/

ID	Statement	Status	Source
C-456	The external 256D associativity CSV (256D_Cayley-Dickson_Basis_Properties.csv) incorrectly claims all 125 tested...	Refuted	‘crates/algebra_core/
C-457	The Cayley-Dickson nested-tuple format uses a binary tree representation where (0, 0) at tree depth k represents 2ˆk...	Verified	‘crates/algebra_core/
C-458	Codebook parity (Thesis A): All lattice codebook points at dims 256, 512, 1024, 2048 satisfy: (a) coordinates in {-1,...	Verified	‘crates/algebra_core/
C-459	Lattice filtration nesting (Thesis B): Lambda_256 is a strict subset of Lambda_512, which is a strict subset of...	Verified	‘crates/algebra_core/
C-460	Prefix-cut characterization (Thesis C): Each filtration transition (2048-¿1024, 1024-¿512, 512-¿256) is a lexicographic...	Verified	‘crates/algebra_core/
C-461	Lambda_32 (the 32-row alignment) is a ”pinned corner” of Lambda_256: all 32 points have first 4 coordinates equal to...	Verified	‘crates/algebra_core/
C-462	XOR partner law (Thesis E): In cross-assessor pairs at dim=64, each pair index i has unique partner i XOR 4 in the...	Verified	‘crates/algebra_core/
C-463	Parity-clique law (Thesis F): ZD adjacency at dim=16 decomposes into K_4 union K_4 by parity of the lower basis index....	Verified	‘crates/algebra_core/
C-464	Spectral fingerprints (Thesis G): Adjacency graph motif classes are distinguishable by spectral invariants (eigenvalue...	Verified	‘crates/algebra_core/
C-465	Scalar shadow action (Thesis D): For each lattice point, the scalar shadow pi(b) = signum(sum(coords)) maps to {-1, 0,...	Verified	‘crates/algebra_core/
C-466	Multiplication coupling lacuna (Thesis D gap): The multiplication-mode lattice coupling rho(b) in GL(8,Z) is...	Theoretical	‘docs/ROADMAP.m
C-467	Null-model identity (Thesis H): For Euclidean-distance ultrametric fraction, RowPermutation and RandomRotation are...	Verified	‘crates/stats_core/sr
C-468	Emanation table (dim=16): The 15x15 signed product table has 84 zero-divisor-marked cells (42 primitive assessor pairs...	Verified	‘crates/algebra_core/
C-469	Strutted emanation table (dim=16): For each strut constant S in {1..7}, the 6x6 strutted ET correctly partitions...	Verified	‘crates/algebra_core/
C-470	Oriented Trip Sync: Every sedenion box-kite admits at least one PSL(2,7) orientation where the de Marrais shorthand...	Verified	‘crates/algebra_core/
C-471	Signed adjacency graph: The strutted ET encodes a signed graph where DMZ cells have sign +1 (same-slope coupling) or -1...	Verified	‘crates/algebra_core/
C-472	Lanyard state-machine traversal: Face-level lanyards are extracted as state-machine traversals of the signed graph....	Verified	‘crates/algebra_core/
C-473	Delta transition function: Each S0 in {1..7} has exactly 3 strut pairs {u,v} with u XOR v = S0, u ¡ v, u,v in...	Verified	‘crates/algebra_core/

ID	Statement	Status	Source
C-474	Brocade normalization: Each BK has exactly 4 brocade relabelings (one per O-trip in its L-set). Each central trip is a...	Verified	‘crates/algebra_core/
C-475	Sail-loop partition (automorpheme duality): The 28 O-trip sails across all 7 BKs partition into exactly 7 loops of 4...	Verified	‘crates/algebra_core/

B Research Insights

B.1 I-001: Macquart Relation Fills the Comoving Distance Gap

Date: 2026-02-06 **Status:** verified **Claims:** C-071

The Macquart relation connects FRB dispersion measures to redshift via integrated baryon density: DM_cosmic(z) = 935 * integral (1+z')/E(z') dz'. Bisection inversion (DM-¿z) converges in ˜27 iterations. Foundation for comoving distance in ultrametric analysis.

B.2 I-002: Ultrametric Structure Lives in Representations, Not Scalars

Date: 2026-02-06 **Status:** verified **Claims:** C-071

C-071 (FRB DMs exhibit p-adic ultrametric structure) definitively refuted using raw DM values. Ultrametricity is a property of hierarchical organization, not scalar distributions. This motivated five new analysis directions testing multi-attribute encodings, temporal cascades, and transformed coordinate spaces.

B.3 I-003: Existing Rust Crate Ecosystem for Cosmological Analysis

Date: 2026-02-06 **Status:** verified **Claims:** none

Identified key crates preventing reimplementation: kodama 0.3.0 (dendrograms), kiddo 5.2.4 (k-d trees, AVX2), fitsrs 0.4.1 (FITS), rustfft 6.4.1 + realfft 3.5.0 (FFT), votable 0.7.0, satkit 0.9.3. Notable gaps requiring custom implementation: cophenetic correlation, Baire metric, local ultrametricity (Bradley 2025), KDE.

B.4 I-004: Kodama Dendrogram and Real Observational Cosmology Infrastructure

Date: 2026-02-06 **Status:** verified **Claims:** C-200, C-201, C-202, C-203, C-204, C-205, C-206, C-207, C-208, C-209, C-210

Kodama returns Dendrogram with Step{cluster1, cluster2, dissimilarity, size}; cophenetic distance c(i,j) = dissimilarity at first merge, enabling cophenetic correlation. Also: first real-data joint fit of Lambda-CDM vs bounce cosmology using 1578 Pantheon+ SNe + 7 DESI DR1 BAO bins. Delta BIC = +7.37 favoring Lambda-CDM. Critical data quality fix: BGS/QSO bins are isotropic-only (not anisotropic).

B.5 I-005: Ultrametric Structure is Radio-Transient-Specific (Preliminary)

Date: 2026-02-06 **Status:** superseded **Claims:** C-437, C-442

Initial 7-catalog survey with 5K subsampling found only FRB/pulsar catalogs showing significant ultrametric excess. SUPERSEDED by I-011 (GPU 10M-triple sweep): the 5K subsampling destroyed Hipparcos galactic signal, making the conclusion too narrow. The ISM-mediation hypothesis for radio transients remains valid.

B.6 I-006: Motif Census Scaling Laws (dim=16..256)

Date: 2026-02-06 **Status:** verified **Claims:** C-126, C-127, C-128, C-129, C-130

Exact scaling laws for Cayley-Dickson zero-divisor box-kite structure across 5 doublings: n_components = dim/2 - 1, nodes_per_component = dim/2 - 2, n_motif_classes = dim/16, n_K2_components = 3 + log2(dim), K2 part count = dim/4 - 1. NO octahedra beyond dim=16.

B.7 I-007: Kerr Geodesic Integrator Verification Summary

Date: 2026-02-06 **Status:** verified **Claims:** C-028

u=1/r regularized Kerr geodesic integrator (Dopri5, Mino time) passes: potential non-negativity, circular photon orbit at 3M, near-horizon infall, a=0.998 stability, r=500 large-distance, shadow area pi*27, asymmetry at a=0.9, coordinate/Mino time monotonicity. Hamiltonian constraint inaccessible from dense output.

B.8 I-008: Cross-Domain Ultrametric Analysis (5K Subsampling)

Date: 2026-02-06 **Status:** superseded **Claims:** C-437

9-catalog ultrametric fraction test with 5K subsampling: only CHIME/FRB and ATNF pulsars pass. Hipparcos at null baseline (p=0.438). SUPERSEDED by I-011: GPU sweep with 10M triples shows Hipparcos 48/114 significant at BH-FDR¡0.05. The 5K subsampling destroyed the galactic spatial hierarchy signal.

B.9 I-009: Elliptic Integral Crate Eliminates Carlson Port

Date: 2026-02-06 **Status:** verified **Claims:** none

The ellip crate (1.0.4, BSD-3-Clause) provides all 5 Carlson symmetric forms (RF, RD, RJ, RC, RG) plus Legendre complete/incomplete integrals (K, E, Pi, D, F). Eliminates need to hand-port Carlson from C++ Blackhole codebase. Tested against Boost Math and Wolfram reference values.

B.10 I-010: nalgebra 0.33/0.34 Version Split Blocks Autodiff

Date: 2026-02-06 **Status:** open **Claims:** none

num-dual 0.13.2 (autodiff via dual numbers) requires nalgebra 0.34, while workspace is pinned to 0.33. Decision: defer num-dual, use closed-form Christoffels for known metrics (Schwarzschild, Kerr, Kerr-Newman). num-dual needed only for generic connection computation on arbitrary metrics.

B.11 I-011: GPU Ultrametric Sweep (9 catalogs)

Date: 2026-02-07 **Status:** verified **Claims:** C-436, C-437, C-438, C-439, C-440

10M triples x 1000 permutations x RTX 4070 Ti via cudarc 0.19. 82/472 tests significant at BH-FDR¡0.05 across 7/9 catalogs. Hipparcos: 48/114 (galactic hierarchy), CHIME/FRB: 8/8 (ALL subsets), GWOSC GW: 4/66 (chirp_mass+q). Supersedes I-008 (5K subsampling bias).

B.12 I-012: The Pathion Cubic Anomaly

Date: 2026-02-07 **Status:** verified **Claims:** C-445, C-446, C-447, C-448

dim=32 (pathion) 8/7 motif split requires degree-3 GF(2) polynomial, NOT degree 4. Degrees 1,2 fail. This proves the polynomial controlling motif-class partition is genuinely cubic.

B.13 I-013: The Hierarchy Fingerprint Theorem

Date: 2026-02-07 **Status:** verified **Claims:** C-441, C-442, C-443, C-444, C-449

Ultrametric fraction test is a genuine hierarchy fingerprint: catalogs with known hierarchical structure (Hipparcos proper motions, CHIME DM) show strong signal, while isotropic catalogs (Fermi GBM GRBs) show no signal. The test discriminates physical hierarchy from noise.

B.14 I-014: Cayley-Dickson External Data Cross-Validation

Date: 2026-02-07 **Status:** cross-validation-complete **Claims:** C-450, C-451, C-452, C-453, C-454, C-455, C-456, C-457

Cross-validated 68 external files against Rust integer-exact computations. Strut table: VERIFIED (C-454). E8 connection: REFUTED (C-455). 8D lattice embedding: VERIFIED at 256/512/1024/2048 (C-452, C-453).

B.15 I-015: Monograph Theses Verification – Lattice Codebook Filtration

Date: 2026-02-08 **Status:** verified **Claims:** C-458, C-459, C-460, C-461, C-462, C-463, C-464, C-465, C-466, C-467

8 monograph theses (A-H) verified: parity constraints, nesting, prefix-cut transitions, scalar shadow, XOR partner, parity-clique, spectral fingerprints, null-model. S_base = 2187 = 3ˆ7.

B.16 I-016: De Marrais Emanation Architecture

Date: 2026-02-08 **Status:** verified **Claims:** C-468, C-469, C-470, C-471, C-472, C-473, C-474, C-475

Implemented L1-L18 emanation layers from de Marrais construction. DMZ = sign-concordance (12 edges/BK), sail-loop = Fano incidence. Oriented Trip Sync universal across all 7 BKs. 113 tests, 4400 lines.

References

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